

Decentralized Exchange: The Uniswap Automated Market Maker

Alfred Lehar*
Haskayne School of Business
University of Calgary

Christine A. Parlour†
Haas School of Business
UC Berkeley

Journal of Finance forthcoming
November 22, 2023

Abstract

Uniswap is a system of smart contracts on the Ethereum blockchain and is one of the largest decentralized exchanges with a liquidity balance worth up to 4 billion USD and daily trading volume of up to 7 billion USD. It is a new model of liquidity provision, so called automated market making. For this new market form, we characterize equilibrium in the liquidity pools. We collect and analyze data on all 95.8 million Uniswap interactions from its inception to December 2022. We provide evidence that the liquidity pools are stable. We compare this automated market maker to a centralized limit order book (Binance) and document absence of long-lived arbitrage opportunities, and show conditions under which the AMM dominates a limit order market.

*Corresponding author, email: Alfred.lehar@haskayne.ucalgary.ca, Tel: (403) 220 4567. Alfred Lehar is grateful to the Canadian Securities Institute Research Foundation and the Fintech Dauphine Chair in partnership with Mazars and Crédit Agricole CIB for financial support. Christine Parlour is grateful for a Norwegian Finance Initiative Grant.

†email: parlour@berkeley.edu. We thank Andrea Barbon, Agostino Capponi, Dmitry Chebotarev, Arthur Gervais, Anand Goel, Terrence Hendershott, Jaisun Li, William Mann, Katya Malinova, Andreas Park, Daniel Rabetti, Ioanid Rosu, Gustavo Schwenkler, Fahad Saleh, Donghwa Shin, Andre Stenzel, Bart Yueshen, and Haoxiang Zhu, for helpful comments, and seminar participants at the BIS, Insead, Penn State, Rochester, University of Copenhagen, University of Florida, 2021 Central Bank Conference on the Microstructure of Financial Markets, 2021 Central Bank Research Association conference, 6th RiskLab Bank of Finland, 2021 China International Conference in Finance, 6th Southern Denmark University Finance Workshop, 2021, NFA 2021, NBIM-Columbia Research Conference, UWA BlockChain and Crypto conference, FRA 2021, AFA 2022, FIRS 2022, Blockchain Technology Symposium, 2021, SNB-CIF Conference on Cryptoassets and Financial Innovation, 2021 FinteqC conference, 2021 Conference on Financial Innovation, Tokenomics 2021, MaCCI Mannheim 2021, 2021 SAFE Microstructure conference, 5th Shanghai-Edinburgh-London Fintech Conference 2021, 2021 Global AI Finance Research Conference, 2021 Collier School of Management Blockchain Research Institute Conference, EFA 2022, and Protocol Labs

Keywords: Decentralized Exchanges, Blockchain, Decentralized Finance

1 Introduction

Uniswap is a decentralized exchange that launched in November 2018. One of the striking features of this exchange is that, instead of a centralized limit order book, it pioneered a novel form of liquidity provision, the so-called automated market making. In such markets, liquidity providers do not explicitly set prices, but rather passively supply liquidity. Price impact for liquidity demanders is calculated mechanically. In spite of this, Uniswap has been very successful. To date, committed liquidity supply tops 4 billion USD across multiple cryptocurrencies and facilitates transactions up to 7 billion USD per day.

In an automated market maker (AMM) such as Uniswap, each asset pair comprises a distinct pool or market. Agents supply liquidity by adding the pair in proportion to the existing pool. Agents demand liquidity by adding one asset and removing the other. The relative proportion of the two traded assets determines the average price paid, and is calculated according to a predetermined downward sloping, convex relationship. This is referred to as a bonding curve. It is surprising that a market in which liquidity suppliers do not explicitly control prices has both viable liquidity supply and is attractive to liquidity demanders.

To analyze this new market form, we collect a novel, detailed data set of 95.8 million interactions with the Uniswap family of smart contracts. Our observations are at block frequency, and include the ordering sequence within each block. Using these data we can identify all flows into and out of 105,098 liquidity pools – that is, changes in liquidity

supply as well as all the token trades. Thus, we can trace how liquidity is both supplied and demanded for each set of asset pairs and the payoffs to participants.

To structure our analysis, we focus on three key differences between an AMM and a limit order market. First, in the AMM, liquidity suppliers are not in competition as all payoffs are shared pro-rata. In contrast, in a limit order book, strategic liquidity suppliers actively compete with each other – the costs and benefits of supplying liquidity are individual to each liquidity supplier. Second, in the AMM, price impact is determined by the bonding curve and is perfectly predictable given the size of the liquidity pool and size of the incoming order. By contrast, in the limit order market, liquidity suppliers choose to post at a price that maximizes their profits and may even withdraw liquidity strategically. Third, in the AMM, liquidity provision and price discovery are separate and provided by different agents who collect different rewards. Liquidity suppliers in the AMM rely on outside arbitrageurs to set the correct price at which the liquidity is provided. They receive trading fee revenue while arbitrageurs get the trading profits.

We first characterize the effect of these three key differences in liquidity provision for an asset whose value is volatile. Risk neutral liquidity suppliers, a liquidity demander and an arbitrageur all interact. Arbitrageurs are strategic traders who profit from stale prices. Liquidity suppliers may be adversely selected as liquidity is posted before any potential asset innovation. We compare strategic liquidity provision in a stylized limit order market to an AMM.

In the stylized limit order book market, competing liquidity suppliers post prices to trade off adverse selection risk against profitable liquidity supply. Of course, if two liquidity suppliers are competing in the same market, each earns zero in expectation. Because of this, each liquidity supplier has an incentive to invest to find trading opportunities in which he does not have to compete. This captures the idea that liquidity is cheap if two suppliers are competing on price. However, if they compete on other dimensions such as speed, the cost of liquidity may increase.

In the AMM, a liquidity supplier deposits a pair of assets into a pool in exchange for a liquidity token. On redemption, the liquidity token returns both assets but in their current proportion. The direct payoff to supplying liquidity is in the form of fees, and the indirect cost of supplying liquidity is the loss of committed capital if an arbitrageur has traded against the pool. This occurs if the relative price of one of the asset shifts, and an arbitrageur finds it profitable to buy the underpriced asset. The arbitrageur effectively rebalances the liquidity suppliers' portfolio at disadvantageous terms.

Our key insight is that the equilibrium size of a pool balances the fee revenue against this “picking off” risk. Specifically, equilibrium is reached through a change in the size of a pool rather than a change in price because larger pools mechanically have a smaller price impact. If picking off risk is high, the equilibrium size of the pool is small because small pools have a higher impact for the same size order. So, faced with a small pool, a rational arbitrageur optimally restricts their trade size which mitigates the extent to which liquidity suppliers' portfolios are adversely rebalanced. A contribution of our paper is thus characterizing the equilibrium size of the liquidity pools as a function of asset fundamentals. Understanding theoretically and empirically the assets for which AMMs are effective is important as more venues experiment with this new market form.

Because of differences in market design between a limit order book and an AMM, gains from trade are split differently across the market participants. In the limit order market, liquidity suppliers retain all the price impact revenue from supplying liquidity, whereas in the AMM, arbitrageurs obtain this benefit while AMM liquidity suppliers only earn the fees. This differentially affects agents' incentives to supply liquidity on either the AMM or in a limit order market. Further, in the limit order market, the fact that there can be both competition that decreases transaction costs and competition that increases transaction costs means that it does not always dominate the AMM for the liquidity trader. Indeed, for assets that have lower volatility (and hence adverse selection) the AMM can be more effective (i.e., is cheaper) at providing liquidity. This observation depends on the fundamental parameters of the tokens traded.

Our extensive data allow us to make four robust empirical observations. First, we document an equilibrium pool size, consistent with our characterization: pools are larger the lower the volatility of the token price, and the higher the level of liquidity trade. Second, we demonstrate that the AMM provides stable liquidity in comparison to a limit order market during an extreme market event. As we document, during extreme market events, both trading fee revenue and gas fees increase. Thus, liquidity providers obtain more revenue and face higher costs to withdraw during market turmoil. Both channels stabilize liquidity supply. Third, consistent with our model, we find that price impact on Uniswap is small with low volatility, while price impact on Binance is higher and exhibits high volatility. Consistent with the predictions of our model we show that price impact on Uniswap is smaller relative to Binance when token volatility is low, for medium trade sizes, and when noise trading is high. Interestingly, in March 2022, Binance integrated Pancake Swap, a Uniswap clone, into the Binance ecosystem and so made their trading venue more similar to an AMM. In the data, we subsequently observe a dramatic drop in observed price impact on Binance as their trading system moved closer to an AMM. Fourth, we document that gas fee differentials between centralized exchanges and decentralized exchanges are negligible for traders who want to keep custody of their tokens.

There is a large literature on liquidity provision in limit order markets. Since Glosten (1994), the efficiency of the limit order book in supplying liquidity has been widely accepted. Most modern markets operate as a form of an open electronic limit order book. Thus, the success of the automated market maker is surprising. In Glosten (1994), the payoff to the marginal liquidity provider at any price balances price concession from liquidity demanders against “picking off risk.” Thus, no competing exchange could offer liquidity at that price and still make profits. As we show, in AMMs, due to pro rata sharing of profits, equilibrium is driven instead by the average liquidity provider. Further, liquidity providers trade off trading fees against a type of “picking off” risk. The size of the pool adjusts, and the bonding curve shifts out until the market reaches equilibrium.

As noted, the payoffs to liquidity provision are shared pro rata in automated market

makers. Although time priority (or first in, first out queuing) is standard practice in most equity markets, there has been little research to determine the optimality of this precedence rule: Glosten (1998) is a rare exception. In this paper, he shows that pro rata rationing changes the marginal payoff to liquidity provision and hence the posted volume. *Ceteris paribus*, we expect pro rata rationing to increase liquidity supply. Haynes and Onur (2020), using a natural experiment from the Treasury Futures market, find that under pro rata rationing, while order sizes and profitability of later submissions are higher, price efficiency is lower. We note that in contrast to equity markets, option markets have long used pro rata rationing. First, to allocate marketable orders across market makers and second to allocate early exercise assignments across writers. Hao, Kalay, and Mayhew (2009) document such rationing in options markets.

More recently, the rise of high frequency traders has generated research into competition that does not lead to cheaper liquidity. Biais, Foucault, and Moinas (2015) present a nuanced view of the effect of speed on market competition as it generates both positive and negative externalities. Empirically, Brogaard, Hendershott, and Riordan (2019) examine the limit orders submitted by HFT on a Canadian exchange. They document high limit order submission and cancellation (95% of the message traffic), which is consistent with strategic liquidity provision.

A few papers have analyzed the theoretical properties of constant function market makers. In a general framework, Angeris and Chitra (2020) show how this class of mechanisms can reflect “true” prices. They also provide a bound on the minimum value of assets held by such an automated system. These two concepts are related because of the increasing price impact faced by a potential arbitrageur. Further, Angeris, Kao, Chiang, Noyes, and Chitra (2021) presents a more specific analysis of Uniswap, while Park (2021) points out blockchain related costs inherent in AMM’s. Aoyagi (2020) characterizes the effect of information asymmetry on these types of markets and shows that the equilibrium liquidity supply size is stable. More recently, Ito and Aoyagi (2021) present a theory model in which an informed trader chooses between a centralized ‘market maker’ market

and an automated market maker. By contrast, we characterize the endogenous liquidity supply for each set of markets and we solve for the optimal venue for a liquidity trader. We also discuss why, in general, the tokens traded on a centralized exchange are not substitutes for tokens traded on a custodial (centralized) exchange.

Most closely related to our work is Capponi and Jia (2021). They present a model and test of an AMM with a focus on competition among arbitrageurs. This competition allows them to consider the joint determination of gas fees and pool size. By contrast, our focus is on the comparison of a limit order market with AMM as markets for liquidity. In particular, we view the gas fees as an important pre-commitment not to withdraw liquidity from the AMM. This allows us to focus on the tradeoff between liquidity fees and adverse selection. We also note that our data are more extensive. A more recent working paper by Barbon and Ranaldo (2021) compares transaction costs and price efficiency on various decentralized exchanges and Binance for five different token pairs. They conclude that due to the fixed cost of gas fees, trading costs on the DEX are roughly comparable to the CEX. Our detailed blockchain data allows a much richer investigation across multiple tokens.

1.1 Detailed Description of Constant Product, Automated Market Making

In this subsection we describe the market making mechanics on Uniswap V2 for readers unfamiliar with the protocol. We present an additional numerical example in Appendix E. Our model is detailed in Section 2 below.

Providing Liquidity: Each swap pool comprises a pair of cryptocurrencies. Most frequently, as we document below, one of the currencies is ETH, the native cryptocurrency on the Ethereum Blockchain. Expositionally, we typically use ETH as the numéraire, and refer to the other generic coin as the ‘token.’ An agent wishing to provide liquidity to their

preferred pool deposits equal amounts of ETH and the token into the pool. The deposit ratio of ETH to token is determined by the existing ratio in the pool, which implicitly defines the ETH price of the token.

An agent who makes such a deposit receives a proportional amount of a liquidity token. This third token is specific to the pool and represents an individual liquidity provider's share of the total liquidity pool. As trade ensues, the value of the liquidity pool may rise or fall. Liquidity providers can redeem their liquidity tokens at any time and get their share of the current liquidity pool paid out in equal values of ETH and tokens. Changes in the composition of the pool from the time a liquidity token is minted to when it is cashed in, potentially constitute adverse selection. However, providing liquidity is potentially profitable because each trade generates a fee of 30bps which is redeposited into the pool. This fee is the same across all pools and hard coded as part of the protocol.

Consummating Trade: Suppose a trader wishes to buy the token. In this case, they will deposit ETH into the pool, and withdraw the token. Conversely, if they wish to sell the token, they deposit it into the pool and withdraw ETH as payment. The amounts that they have to deposit or withdraw are calculated from the bonding curve which is illustrated in Figure 1.

Suppose there are E_0 ETH and T_0 tokens in the pool and a trader wishes to sell the token. They deposit the amount for sale, $T_1 - T_0$, into the pool. In return, they receive $E_0 - E_1$ and the amount of ETH in the pool drops to E_1 .

If the seller was a liquidity trader, the post trade price in the pool ($\frac{E_1}{T_1}$) is now too low, and a potential arbitrageur would enter the market and trade in the opposite direction to return the ratio of ETH to tokens to equilibrium. Note that the liquidity supplier does not benefit from the price impact paid by the liquidity demander – this benefit accrues to the arbitrageur.

The terms of trade are such that for any post trade quantities before any fee revenue T_1 ,

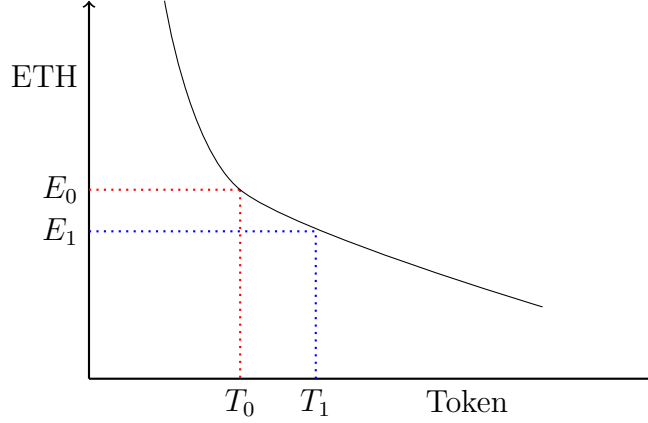


Figure 1. A bonding curve. From an initial amount of ETH and Tokens of E_0 and T_0 respectively, a trader deposits $T_1 - T_0$ tokens (sells) in exchange for $E_0 - E_1$ ETH. the price impact of this trade is determined by the bonding curve.

E_1 satisfy

$$k := T_1 \cdot E_1 = T_0 \cdot E_0. \quad (1)$$

In other words, the product of the token and ETH quantities is always on the bonding curve.

For each pool, the constant k , depends on the amount of liquidity that has been deposited in the pool up to this point. Further, if more liquidity is posted, the constant increases. (The bonding curve shifts out along a ray.) This is the mechanism through which the market equilibrates.

Assessing Liquidity Fees: The previous clarifies the terms of trade absent the liquidity fee. Of course, remuneration is important for the liquidity providers. To see how the fee affects trades and prices, suppose that an agent wants to trade e ETH in exchange for tokens. The exchange collects a percentage fee τ , which benefits liquidity holders. Thus the effective amount of ETH that trades is $(1 - \tau)e$. This leads to a post trade, but before fee revenue liquidity pool balance of $E' = E + (1 - \tau)e$. Following the logic of the bonding curve (Equation 1), the post trade token balance must be

$$T' = \frac{T \cdot E}{E'} = \frac{T \cdot E}{E + (1 - \tau)e}. \quad (2)$$

The smart contract which executes the trade accepts the e ETH and returns the difference between the pre and post trade token balances. Or, the amount of token t that the trader receives is given by

$$t = T - T' = \frac{(1 - \tau)eT}{(1 - \tau)e + E}. \quad (3)$$

Therefore, the terms of trade expressed in ETH/token is given by

$$p^{tot} = \frac{e}{t} = \frac{e}{T} + \frac{E}{(1 - \tau)T}. \quad (4)$$

The terms of trade have a natural interpretation as a spread. Suppose that the fundamental value of the token denominated in ETH is p_0 . If the pool is in equilibrium then $p_0 = \frac{E}{T}$. The liquidity fee generates what is essentially a tick size that is distinct from the volume-induced price impact that the trader pays when he moves long the bonding curve, then

$$\lim_{e \rightarrow 0} \frac{p^{tot}}{p_0} = \frac{ET}{ET(1 - \tau)} = \frac{1}{1 - \tau} \quad (5)$$

That is, when buying tokens, traders have to pay a fixed spread of $\frac{1}{1 - \tau}p_0$. Similarly for token sales traders have to pay a fixed spread of $(1 - \tau)p_0$.

Pool size: The price that a trader gets is determined by the bonding curve and the volume of posted liquidity. In particular, the price impact of a marginal increase in the order is $\partial p / \partial e = 1/T$. As the liquidity pool grows, the price impact of a fixed order size decreases. Thus, understanding the payoff to liquidity provision and hence the size of each liquidity pool is an important determinant of AMM market quality.

Figure 2 presents an example of an ‘orderbook’ that an incoming trader might face. The blue line is for a small pool and the orange line for a larger pool. Because Uniswap has a

unique mapping of trading quantity to price, the graph shows the exact amount that is traded at any price. The spread or fixed cost of trading is evident in the interval around the mid-price of 10 for which no quantities can be bought. Clearly, larger orders face worse marginal prices as they would in a typical limit order book.

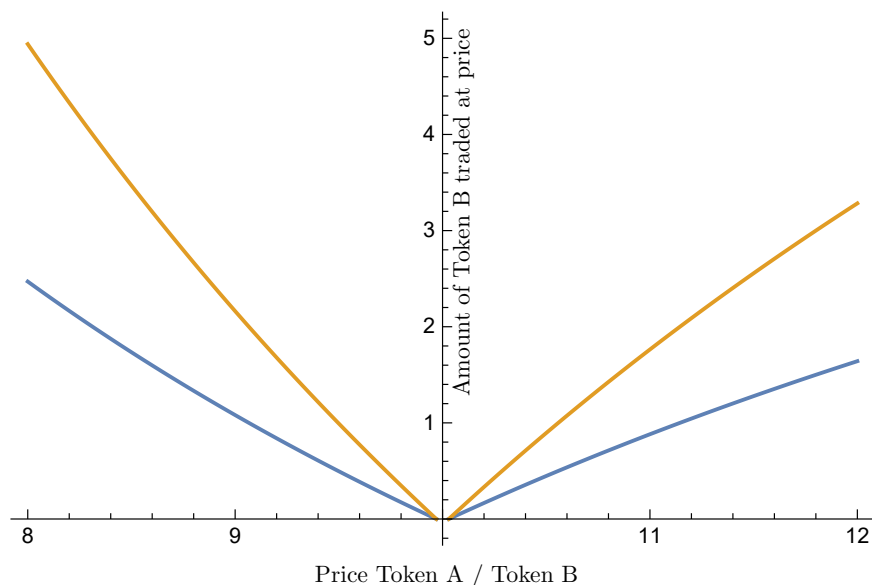


Figure 2. Uniswap orderbook depth The graph shows how many Token B could be bought or sold at a given price for a large (orange) and small (blue) liquidity pool, respectively. The parameters are: $\tau = 0.003$ and $T = 20, E = 200$ for the large pool, and $T = 10, E = 100$ for the small pool.

2 Framework

Consider a market for one asset, with current value p_0 . With probability α there is an innovation and the asset is equally likely to jump up or down to $p_0 + \sigma$ or $p_0 - \sigma$ respectively, else the asset value remains p_0 . Thus, $\frac{\sigma}{p_0}$ is the proportional innovation. A potential arbitrageur monitors the market and trades whenever profitable, otherwise a liquidity trader, who trades a fixed quantity q , arrives. The liquidity trader is equally

likely to buy or sell, at any price $p \in [p_0 - \sigma, p_0 + \sigma]$. At the extreme prices a trading crowd stands ready to execute orders.

2.1 Limit order market

There are two rational, deep pocketed, liquidity suppliers who potentially enter the market before the passive trader and supply liquidity that optimally trades off the surplus they can extract from the liquidity trader against the possibility of being “picked off” by an arbitrageur. We focus on the case of two liquidity suppliers as it is the minimum required for competition. These rational liquidity suppliers can privately monitor the market to increase the probability that they will be a monopolist liquidity supplier. The symmetric monitoring cost is $I(\gamma) = a\gamma^2$. If liquidity supplier i chooses γ_i and liquidity supplier j chooses γ_j then nature assigns i to be the sole liquidity supplier with probability $\gamma_i(1 - \gamma_j)$. This private monitoring captures the idea that liquidity suppliers have an incentive to compete in many different ways, such as co-location or speed.

To simplify the exposition, in the text we describe the case where the informed trader buys, i.e., if there is an innovation the asset value jumps up. The case where the informed trader sells is symmetric. We characterize symmetric equilibria.

The sequence of events in the limit order market is as follows: First, each liquidity supplier chooses their level of private investment. Second, nature determines if there is one or two limit order submitters, and then each limit order submitter posts orders. In the case of competing liquidity suppliers one of them is randomly selected to go first. This captures the importance of time priority. After liquidity is posted, Nature determines the new asset value. If there is no relative value change, the liquidity trader arrives and trades against the best quote or randomizes if indifferent. If there was a relative value change, the arbitrageur trades if it is profitable.

From the above, it follows that liquidity suppliers may be alone or competing in a market.

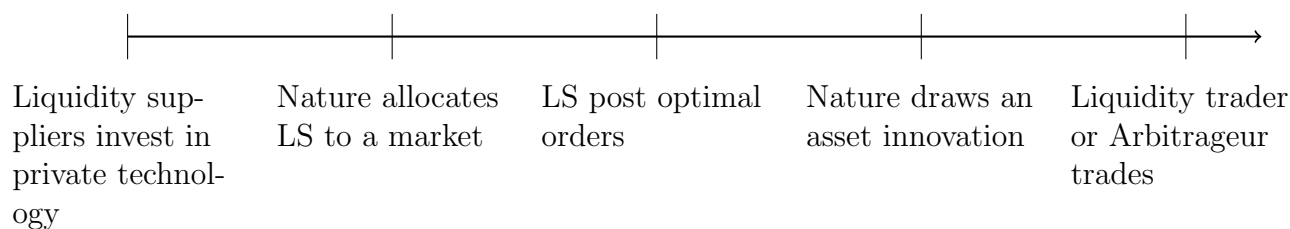


Figure 3. Sequence of Events in the Limit Order Market

We characterize their optimal trading strategies in both cases, and then consider their private investment. The amount that the liquidity trader trades is fixed, q , and so this is also the amount that the liquidity suppliers post.

If a liquidity supplier is alone in the market, then he will always post a sell price of $p_0 + \sigma$. Posting at this high price completely mitigates adverse selection, and at the same time extracts maximal surplus from the passive trader.

Lemma 1 *A monopolist liquidity supplier in the market, will post a sell price of $p_0 + \sigma$, and a buy price of $p_0 - \sigma$ to obtain an ex ante profit of $q(1 - \alpha)\sigma$.*

By contrast, if two competing liquidity suppliers are in the market then the first liquidity supplier rationally anticipates that he will always be undercut unless he offers the competitive price at which he makes zero profit. The second liquidity provider rationally anticipates that the noise trader's order will be filled by the first liquidity provider and he can at best trade with the arbitrageur. He will therefore post a price of $p_0 + \sigma$.

Lemma 2 *If two competing liquidity suppliers are in the market offering orders to sell, then the first liquidity provider will post a sell price of $p_0 + \alpha\sigma$ and a buy price of $p_0 - \alpha\sigma$, while the second liquidity provider will post a sell price of $p_0 + \sigma$ and a buy price of $p_0 - \sigma$. Each competing liquidity supplier makes zero profits.*

A sole liquidity supplier makes positive profits, while those in competition make zero

profits. Increasing private investment makes it more likely that a liquidity supplier makes positive profits.

The logic in support of competitive liquidity provision is that competition leads to lower prices. However, this is only true if suppliers are only competing on price. Given the complexity of electronic trading platforms, liquidity suppliers compete in many different ways. Such competition does not necessarily lead to lower prices and may even be inefficient. In the context of our framework, we capture this through liquidity suppliers' private investment to increase the likelihood that they are a monopolist.

Proposition 3 *Each trader chooses private investment, $\gamma^* = \frac{(1-\alpha)\sigma q}{2a+\sigma q(1-\alpha)}$. The optimal private investment is increasing in the innovation in price, σ and the size of the liquidity trade q , and decreasing in the probability of an innovation, α , and the monitoring cost, a .*

The limit order traders find it profitable to trade with the noise traders, and σ captures the extent to which they can extract surplus from them.

2.2 Automated Market Maker

In the AMM or bonding curve market, liquidity suppliers choose a pool and commit quantities of both ETH and Tokens. The first thing to observe is that liquidity provision is not rivalrous and there is no incentive for private investment as there is in the limit order market. The second thing to observe is that the AMM requires committed capital.

Suppose that investors have each committed E_0 ETH and T_0 tokens. Two identities from the bonding curve will be useful: First, the ratio of ETH to tokens, in equilibrium, is the ETH price of tokens implied by the bonding curve market or

$$\frac{E_0}{T_0} = b_0. \quad (6)$$

We will start our analysis under the assumption that the price in the bonding curve market is equal to the equilibrium price, or $b_0 = p_0$. Second, any transactions must occur along the curve. Specifically, if an amount t tokens and e ether are traded, then

$$E_0 T_0 = k, \quad (7)$$

$$\text{and} \quad (E_0 + e)(T_0 + t) = k \quad (8)$$

where k is the constant of the bonding curve.

In addition, as we indicated in subsection 1.1, any trade is assessed a proportional fee of τ . We simplify the algebra that follows by assuming that liquidity fees do not change the size of the pool, but are placed into a separate account. In reality, liquidity fees are paid into the pool and therefore change the bonding curve constant. The timing of events is shown in Figure 4 below.

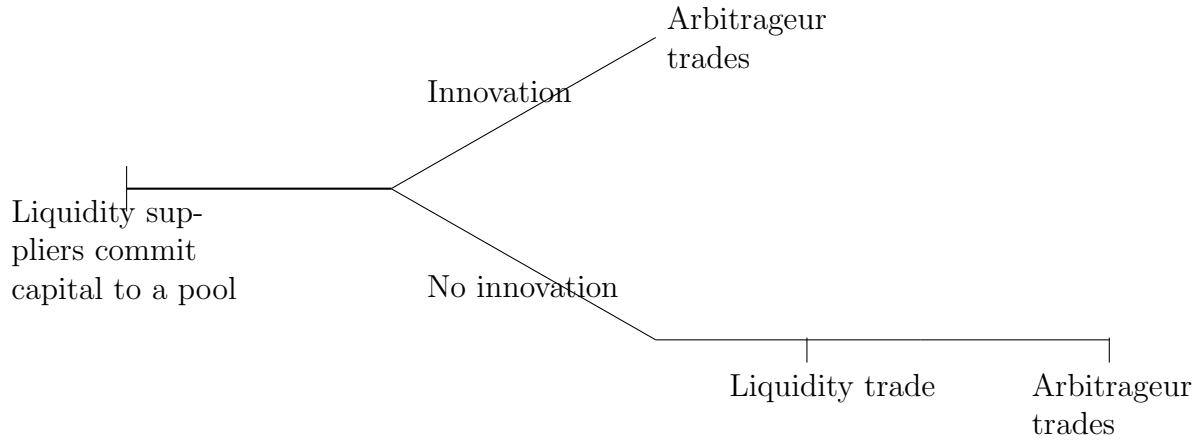


Figure 4. Sequence of Events in the Automated Market Maker

First consider the payoffs to liquidity provision if a liquidity trader arrives. If the liquidity trader buys tokens, they will remove q tokens, and will buy these with the numeraire good, ETH. Thus, they add in e_ℓ^b to the ETH pool. The specific amount of ETH they add is

determined by the bonding curve, so

$$\begin{aligned}(E_0 + e_\ell^b)(T_0 - q) &= E_0 T_0 \\ e_\ell^b &= \frac{E_0 T_0}{T_0 - q} - E_0.\end{aligned}$$

Per the protocol, liquidity providers receive a fee for facilitating this transaction. We model this by assuming that a fee is deposited into a separate account. The fees paid by this noise trader are τe_ℓ^b .

Reversing this trade (so selling tokens to the pool) is profitable for the arbitrageur if $e_\ell^b(1 - \tau) < qp_0$. If τ is sufficiently small, the arbitrageur will add tokens and remove ETH. Under our maintained assumptions, the payoff to liquidity provision is twice the fee paid by the liquidity trader. Or, $2\tau e_\ell^b$. Symmetrically, if the liquidity trader is a seller, they will deposit q tokens and remove $e_\ell^s = E_0 - \frac{E_0 T_0}{T_0 + q}$. The arbitrageur will buy tokens. These two transactions generate a fee revenue of $2\tau e_\ell^s$.

In contrast to a market maker in a limit order market, the AMM cannot adjust its price and has to rely on arbitrageurs to align the quoted price with the fundamental value. Thus, in an AMM, liquidity provision and price discovery are separate and provided by different agents who collect different rewards. The liquidity providers receive the fee revenue while the arbitrageur earns trading profits.

Our formulation makes use of two simplifications for tractability. First, in reality, the arbitrageur faces a different size pool than the liquidity trader as the liquidity fee has been paid into the pool. Second, the arbitrageur will trade slightly less than e_ℓ^b . The overall size of these two effects is negligible and of the order τ^2 .

Lemma 4 *Suppose that the aggregate amount of ETH and Tokens in a liquidity pool are E_0 and T_0 respectively, and τ is sufficiently small. With probability $(1 - \alpha)$, there is a*

liquidity event and the fee revenue for liquidity provision is:

$$2\tau p_0 q \left(\frac{T_0}{T_0^2 - q^2} \right). \quad (9)$$

Now suppose that there was a positive innovation event so that an informed arbitrageur arrives. Since the pricing is deterministic she will trade an amount that maximizes her profit. She will buy t^b tokens and pay e_I^b for them. The ETH payment is pinned down by the bonding curve, which requires that $(T_0 - t^b)(E_0 + e_I^b) = E_0 T_0$ which gives her a profit function of

$$\pi_I^b = (p_0 + \sigma)t^b - (1 + \tau) \left[\frac{E_0 T_0}{T_0 - t^b} - E_0 \right]. \quad (10)$$

Given the convexity of the bonding curve, the optimal trading amount is determined by the first order condition,

$$(p_0 + \sigma) - (1 + \tau) \frac{E_0 T_0}{(T_0 - t^b)^2} = 0, \quad (11)$$

which implies

$$t^b = T_0 - \sqrt{\frac{(1 + \tau) E_0 T_0}{p_0 + \sigma}} \quad (12)$$

$$e_I^b = \sqrt{\frac{E_0 T_0 (p_0 + \sigma)}{(1 + \tau)}} - E_0. \quad (13)$$

We note that this is optimal for the informed arbitrageur only if $t^b \geq 0$, or $\tau \leq \frac{\sigma}{p_0}$, so that the transaction cost is low relative to the innovation.¹

After the innovation, absent informed arbitrage trading, the ETH value of the total sup-

¹If $\tau > \frac{\sigma}{p_0}$, the proportional innovation, then the fees are so high that an arbitrageur would never trade. Liquidity traders would also be better off choosing the limit order market, and the AMM would not be viable.

plied capital would be $E_0 + (p_0 + \sigma)T_0$. Given the informed arbitrage trade, the ETH value of the supplied capital is

$$\begin{aligned} & E_0 + e_I^b + (p_0 + \sigma)(T_0 - t^b) \\ = & \sqrt{E_0 T_0 (p_0 + \sigma)} \left(\frac{2 + \tau}{\sqrt{1 + \tau}} \right). \end{aligned}$$

Therefore, the change in value of supplied capital for liquidity suppliers after an increase in the value of the asset is:

$$\left(\frac{2 + \tau}{\sqrt{1 + \tau}} \right) \left(\sqrt{T_0 E_0 (p_0 + \sigma)} \right) - (E_0 + (p_0 + \sigma)T_0). \quad (14)$$

This change in value corresponds to “picking off” risk, in the sense that the informed arbitrageur rebalances the amount of ETH and Tokens to reflect the value in the wider market. In addition, however, the arbitrageur pays a liquidity fee. Consistent with the previous case, we assume that this is levied on the ETH total, for an amount equal to $\tau \left(\sqrt{\frac{E_0 T_0 (p_0 + \sigma)}{(1 + \tau)}} - E_0 \right)$.

Lemma 5 *Suppose that the aggregate amount of ETH and Tokens provided are E_0 and T_0 respectively, and $\tau \leq \frac{\sigma}{p_0}$. With probability $\frac{\alpha}{2}$, the asset value jumps up and the payoff to liquidity provision is:*

$$\underbrace{\left(\frac{2 + \tau}{\sqrt{1 + \tau}} \right) T_0 \left(\sqrt{p_0 (p_0 + \sigma)} \right) - (T_0 p_0 + (p_0 + \sigma)T_0)}_{\text{Picking off}} + \underbrace{\tau \left(T_0 \sqrt{\frac{p_0 (p_0 + \sigma)}{(1 + \tau)}} - p_0 T_0 \right)}_{\text{liquidity provision}}. \quad (15)$$

A symmetric expression holds for a jump down in the asset value.

Armed with Lemmas 4 and 5 we can determine the overall payoff to liquidity provision for the entire pool and thus the equilibrium size of the pool. The equilibrium pool size

makes liquidity suppliers indifferent between committing capital to the AMM or to their outside option which, for simplicity, we normalize to zero.

Proposition 6 *Suppose that $0 < \tau < \frac{\sigma}{p_0}$, then the equilibrium supply of Tokens is given by*

$$T_0 = q \left[\sqrt{1 + \frac{(1 - \alpha)^2 \tau^2 p_0^2}{\alpha^2 \omega^2}} - \frac{(1 - \alpha) \tau p_0}{\alpha \omega} \right] \quad (16)$$

where $\omega = \sqrt{p_0(p_0 + \sigma)(1 + \tau)} + \sqrt{\frac{p_0(p_0 - \sigma)}{1 + \tau}} - 2p_0$.

Figure 5 illustrates the tradeoff captured in Proposition 6. In this figure, fundamental asset characteristics are fixed. The left panel illustrates trade size as a function of pool size. By assumption, the noise trader (blue) trades a constant amount. Following a noise trade, the arbitrageur trades an identical amount in the opposite direction to move the pool price back to its fundamental value. If there is an asset innovation, the arbitrageur optimizes trade size to maximize profits. The larger the pool, the lower the price impact, and the more the arbitrageur will optimally trade to push the pool price to its new fundamental value. The right panel shows fee revenue (blue) and the adverse selection cost (orange) as a function of pool size. The adverse selection cost of the pool (orange line, left panel) is proportional to the arbitrageur's trade size. Liquidity providers trade off the fee revenue with this adverse selection cost. Fee revenue is slowly growing in pool size as overall trading volume increases because of the arbitrageur.

The parameter restrictions under which Equation (16) characterizes pool size are intuitive. If the payoff to liquidity suppliers is too small, then pools are not sustainable. If the payoff to liquidity provision is too large, then arbitrageurs will not find it lucrative to trade on relative price information, and will also not trade to ensure that the price implied by the pool corresponds to the traded value of the token. Going forward, we assume that these conditions hold because our empirical focus is on the pools which we observe; further as

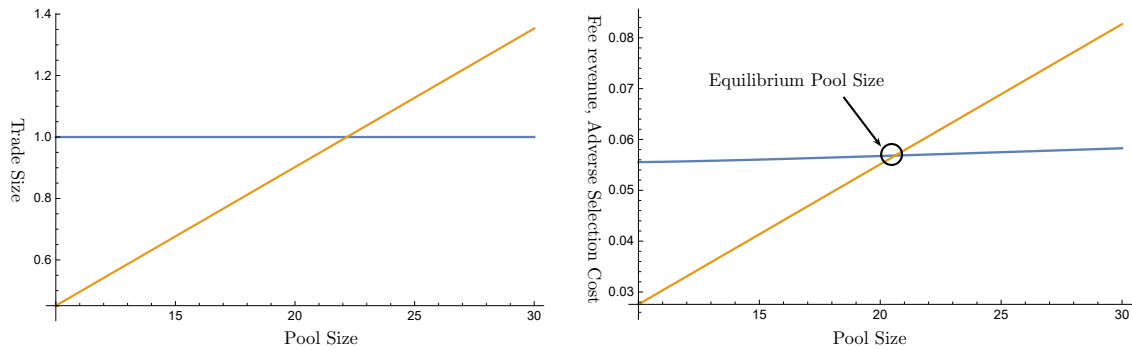


Figure 5. Optimal pool size. The left panel shows the optimal trade size by the arbitrageur upon a price innovation (orange) and the liquidity trader (blue) as a function of pool size. The right panel shows fee revenue (blue) and adverse selection cost (orange) as a function of pool size. The parameters are $P_0 = 10$, $\sigma = 1$, $q = 1$, $\alpha = 0.1$, $\tau = 0.003$

we show in our empirical analysis, the prices implied by Uniswap consistently track traded prices on Binance which suggests that cross-market arbitrage is profitable.

3 Data and stylized facts

Decentralized exchanges (DEX) are smart contracts mostly deployed on the Ethereum blockchain. Users initiate trade as an Ethereum transaction that sends tokens to a smart contract which calls a function to perform the exchange. The smart contract then sends trade proceeds in the form of the appropriate tokens back. Since transactions on Ethereum are atomic, or in other words they either execute completely or fail, there is no settlement risk and users do not have to hand over custody of their digital assets to a third party. In keeping with other DeFi protocols, the source code for many DEXs is public and users can verify that the code is not fraudulent. This also allows them to perfectly predict how the smart contract will perform, and more specifically, the terms of trade if they submit an order.

Uniswap is open source with no owner or operator. It was launched in November 2018 at Devcon 4, and in its first release, Uniswap V1 allowed exchange between any ERC20 tokens against Ether (ETH).² Because Uniswap has no designated operator, V1 pools cannot be deleted from the blockchain and exist in parallel to V2 pools. Uniswap V2 was launched on May 18, 2020. The update allows direct trade of any ERC 20 token pairs. If no pool exists for a specific token pair, it can be freely created by invoking the Uniswap factory contract, and specifying the token for which a new pool should be created. The factory contract will then deploy a new pool for that specific token on the Ethereum blockchain, and enable subsequent trade. As we document below, most V2 pools trade tokens against wrapped Ether (WETH), which is a ERC 20 representation of Ether (ETH).

To collect our data, we obtained a list of all Uniswap V1 and V2 liquidity pools from the original factory contract transactions. Our sample comprises 105,098 individual liquidity pools, consisting of 2,126 V1 pools and 102,972 V2 pools. We matched transactions into and out of these liquidity pools with block-by-block transactions on the Ethereum blockchain. In total we have 94,232,292 transactions on Uniswap from its inception on November 2, 2018 until December 21, 2022. We observe 2,097,032 liquidity injections into a pool, 959,312 withdrawals of liquidity from a pool, and 91,175,948 trades of tokens.

An Ethereum transaction is a set of instructions that are processed as one unit within the same block. Apart from implicit limits on transaction sizes given by Ethereum block size, there is no restriction on how many interactions with Uniswap liquidity pools can be done in one transaction. In our sample 65.62% of Ethereum transactions only have one interaction with one liquidity pool, another 14.13% have two interactions. 5,956 transactions or 0.0063% of the sample have 10 or more interactions with a liquidity pool. One transaction interacted with 141 different liquidity pools.³ The most complex transaction in terms of

²ERC 20 is the most commonly used standard for fungible tokens on Ethereum compatible blockchains.

³See transaction 0xf10dfb01d0b2e13a28e7bde411bcb4321b703a55e96d11487b382effaa8998b7, in which someone swapped for no apparent reason 0.0000001 Ether each with 140 different liquidity pools.

number of swaps in our sample has 228 interactions with 2 different Uniswap liquidity pools.⁴ Uniswap V2 introduced router contracts that can perform complex transactions with one function call. Assume for example that a pool exists that trades tokens A and B, another pool trades tokens B and C, and there is no pool to swap A and C. The router contract can then be instructed to swap A and C by trading through token B. In our sample such a transaction would show up as two separate transactions, one for each of the two pools.

It is important to recognize that our analysis is on liquidity supply on both a limit order market and an AMM. Therefore, we do not consider market access fees – either on the limit order market (e.g., co-location, make-take fees) or on the AMM (e.g., gas fees). The latter are analyzed in Capponi and Jia (2021).

Most pools trade against WETH. To compute volume we take the WETH part of the trade and convert it to USD using Binance minute by minute data. Most of the remaining pools trade against a USD stablecoin. For those pools we convert the amount traded in the stablecoin to USD. For all remaining pools we search for all pools where one of the tokens trades against WETH and convert it using the prices from the pool with the highest volume. More on the network structure of Uniswap can be found in Appendix D.

Table 1 provides an overview of the 10 largest Uniswap V1 and V2 pools by average daily volume in ETH. We exclude pools with less than 5,000 trades and an average poolsize of less than 700 ETH. V1 pools are smaller both in terms of volume as well as in number of trades, mostly because the introduction of V2 coincided with a huge boom in Decentralized Finance and caused most traders and liquidity providers to converge on the new protocol. The largest pool with at least 100 trades in terms of total volume traded is USD Coin (USDC) - Wrapped Ether with an aggregate volume (over all days) of over 43.23 billion USD. The Wrapped Ether - Tether USD pool has the highest number of total trades in our sample – over 4.61 million trades. The “on average” largest pool in ETH is Wrapped

⁴see Ethereum transaction 0xa7c81e54c54e58db88f01a92c3602b0996b82ff424b549cb9e8d918e2614ecf5.

BTC - Wrapped Ether with an average size of around 225,721 ETH.

While some of the pools are very active, many are not: 78,341 pools in our sample have fewer than 100 transactions. Figure 6 shows the number of trades by pool type. With the release of V2 trading activity in V1 declined. We also observe that for V2 pools trading against WETH (orange) dominates direct trading of other tokens (red).

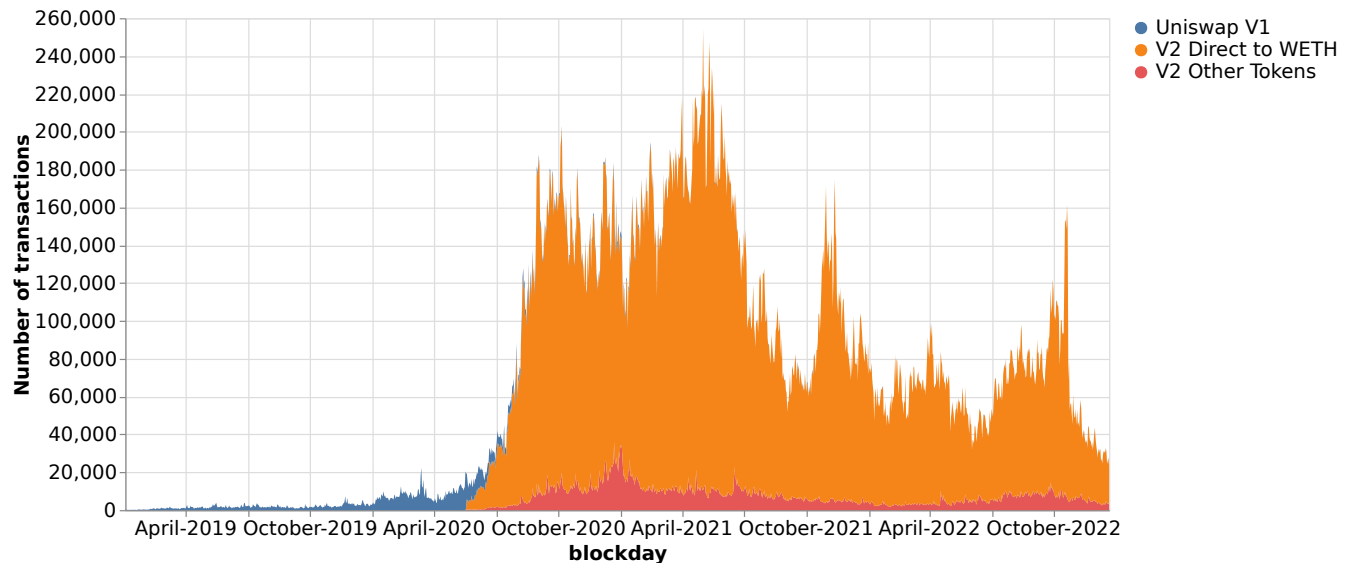


Figure 6. Number of transactions on Uniswap.

Figure 7 illustrates the trading volume per day. Computing volume is not straightforward in bonding curve markets as attackers often deliberately push markets out of equilibrium. The highest volume in our sample is on March 30, 2021 with a volume of 18.34 billion USD. On that day, a trader moved 5.5 billion USD of a token back and forth between her own wallets. On October 26, 2020 there is volume of over 5.5 million ETH or USD 2.1 billion, linked to an attack on Harvest Finance using a flash swap. A more detailed discussion of this incident and implications for Uniswap volume can be found in Appendix C. Use of Flash Swaps varies in our sample. Out of the 960 days when V2 was deployed, flash swaps occurred on 922 days. The median flash swap volume per day was 164,005 USD and the maximum was 17.16 billion USD on March 30, 2021. The first flash swap

Token 1		Token 2		Panel A: Uniswap V2		Num.Tx	Volume (ETH)	Volume (USD)	Pool size (ETH)	Days active
USD Coin	USDC	Wrapped Ether	WETH	4,397.4	35,050	45,027,074	122,236	960		
Wrapped Ether	WETH	Tether USD	USDT	4,868.7	33,751	41,344,518	139,117	946		
Dai Stablecoin	DAI	Wrapped Ether	WETH	1,552.6	16,862	16,854,140	126,240	954		
Uniswap	UNI	Wrapped Ether	WETH	868.9	9,725	9,684,291	47,004	825		
Wrapped BTC	WBTC	Wrapped Ether	WETH	478.0	8,866	9,385,463	225,721	947		
yearn.finance	YFI	Wrapped Ether	WETH	322.3	5,710	2,858,581	25,572	886		
Wrapped Ether	WETH	Ampleforth	AMPL	451.0	5,140	2,734,632	34,048	947		
Fei USD	FEI	Wrapped Ether	WETH	112.8	4,818	12,789,495	128,547	633		
ChainLink Token	LINK	Wrapped Ether	WETH	436.3	3,905	4,373,169	21,055	947		
SushiToken	SUSHI	Wrapped Ether	WETH	318.3	3,878	1,978,673	68,339	847		
Table volumes V1	Ether	Dai Stablecoin	DAI	273.2	1,264	258,767	8,908	1,132		
Ether	ETH	HEX	HEX	113.6	818	174,572	20,633	1,113		
Ether	ETH	USD Coin	USDC	167.5	760	181,431	6,447	1,485		
Ether	ETH	Maker	MKR	89.4	756	144,604	10,249	1,510		
Ether	ETH	Sai Stablecoin v1.0	SAI	122.2	565	107,241	5,030	1,510		
Ether	ETH	LoopringCoin V2	LRC	11.8	545	202,887	771	1,308		
Ether	ETH	Synthetix Network Token	SNX	65.5	359	66,961	3,505	1,230		
Ether	ETH	Synth sETH	sETH	22.7	201	38,153	26,515	1,195		
Ether	ETH	UniBright	UBT	55.2	139	28,975	634	1,185		
Ether	ETH	Pinakion	PNK	25.2	119	36,217	1,536	1,504		

Table 1: **Ten largest exchanges for Uniswap V1 and V2, respectively, sorted by average daily volume (in ETH).** *Number transactions* is the daily average number of transactions, *Volume (ETH)* is the daily average volume in Ether, *Volume (USD)* is the daily average volume in USD, *Pool size (ETH)* is the daily average pool size in Ether, and *Days active* are the days since inception of that pool.

occurred on May 23, 2020.

The largest non-flash swap trade in our sample was on December 17, 2020 when a trader swapped 48,584,947.17 DAI for 342,252.89 WETH worth about USD 220.42 million at the time as part of an attack on the platform Warp finance. On June 18, 2020, a trader swapped 100,000.39 WETH (about USD 23.2 million USD at the time) for 1,695,998.19 UniBomb tokens as part of another exploit.⁵ The median trade size in our sample is 669.95 USD. 6.70% of trades are below 0.01 ETH and 19.21% are below 100 USD. We provide details on our methodology in Appendix C.

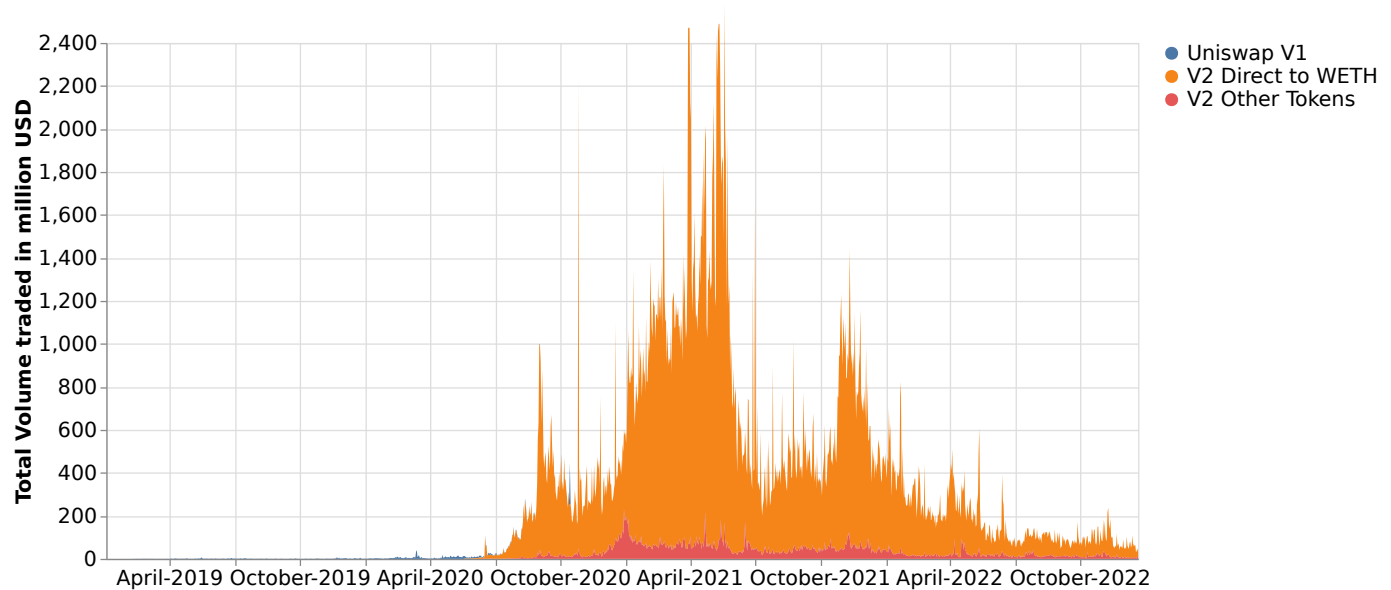


Figure 7. Trading volume on Uniswap The graph shows the daily volume of transactions including flash loans in USD. The graph is capped at 2.4 billion USD.

For our subsequent econometric analysis we purge pools that are very small or were only used for a few days. Specifically, we drop pools with less than 30 trading days and with less than 100 ETH average balance. The reduced sample has 59,606,977 observations in

⁵See transaction 0x0x8492ce3b1ea8ec796471997731e557c057c2fb0a3ade7f9c0477450d53ad4791. Unibomb is deflationary token that burns 1% of the each transaction, thus increasing its value. Somebody seems to have borrowed 100.000 ETH from the lending platform dXdY and converted them to Unibomb. The transaction decreased token supply and the user could reconvert the Unibomb tokens to ETH with a slight gain in price, leaving a profit after the repayment of the loan.

1,525 pools.

4 Liquidity provision

4.1 Pool size

We have demonstrated empirically that there is heterogeneity in pools. The equilibrium pool size in Equation (16), permits us to derive comparative statics.

Proposition 7 *Suppose that the current price is p_0 and the fee is less than the proportional innovation or $\tau < \frac{\sigma}{p_0}$, then the equilibrium size of a liquidity pool is*

- i. Linear in the size of the liquidity trade q .*
- ii. Is decreasing in the size of the innovation, σ .*
- iii. Is decreasing in the innovation probability, α .*

In equilibrium, pool size adjusts so that a pool's per token fee revenue balances the cost from being picked off by arbitrageurs if the pool posts stale prices. When innovations, σ , are larger, the informed arbitrageur will ceteris paribus place larger trades, and losses for liquidity providers will therefore be larger. By contrast, fee revenue from the liquidity trader is independent of movements in the fundamental price. This implies that to increase the expected payoff per unit of liquidity, the pool in equilibrium must be smaller. As the pool shrinks, price impact increases and so the informed arbitrageur places a smaller order. These comparative statics highlight how the AMM reaches equilibrium – as the price impact is a deterministic function of the pool size, the size of the pool adjusts to ensure that the liquidity suppliers receive the opportunity cost of their capital.

The equilibrium pool size also decreases in α , the probability of an innovation because this generates informed trading, i.e. an arbitrageur trading because of an innovation. As

α increases the fraction of liquidity traders decreases and the pool is more likely to be picked off by arbitrageurs. The pool size again shrinks in equilibrium, increasing price impact, and reducing the optimal trade size of the informed trader. Figure 8 shows the poolsize as a function of σ and α for a numerical example. We summarize our empirical prediction below.

Hypothesis 1 *We expect poolsize to decrease in the volatility of the exchange rate of the two tokens and increase in uninformed trading.*

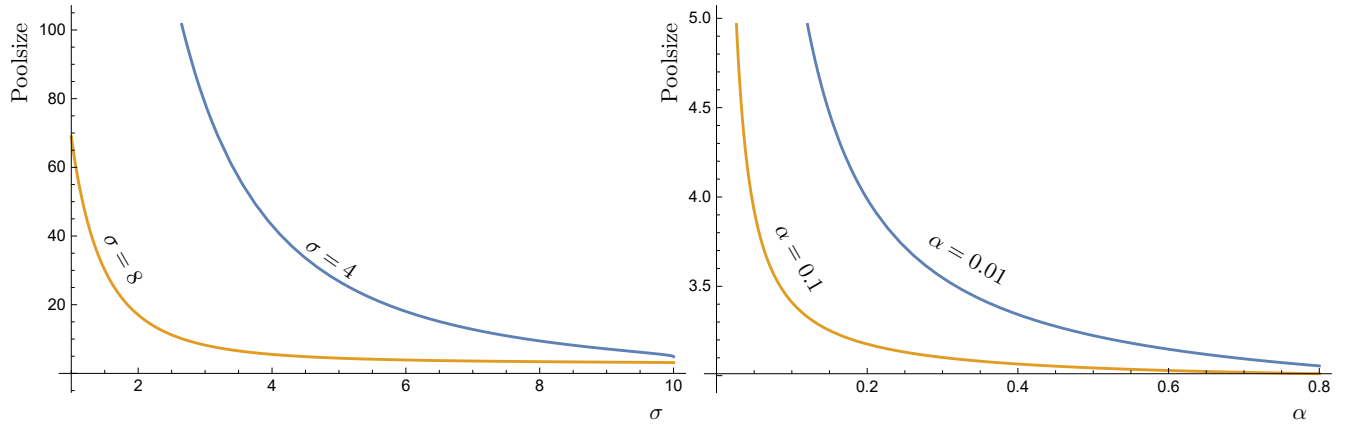


Figure 8. Poolsize as function of innovation size σ and α . Unless otherwise stated the parameters are $\alpha = 0.01, p_0 = 10, \sigma = 4, q = 3, \tau = 0.003$

To test the predictions of Hypothesis 1, we collect daily data on 1,525 pools in our final sample. For the average pool, we observe 702 days, while the median is 750. To test Hypothesis 1 we specify the following regression at a pool-day level:

$$poolsize = a + b_1\sigma_{fx} + b_2\mathbb{1}_{airdrop} + b_3noise\ trading \quad (17)$$

Where *poolsize* is the average daily pool size in USD which is the daily average of poolsizes after each trade converted to USD using minute by minute data. We control for the airdrop of UNI tokens during Uniswap's liquidity minting program as they might have affected

the return for liquidity providers and thus induced a larger pool size.⁶ We set the dummy $\mathbb{1}_{airdrop}$ equal to one for included pools on airdrop days. We control for the daily block by block exchange rate volatility, σ_{fx} , as a measure of adverse selection.⁷ We use three measures for noise trading. First, we use the number of trades on a day in a pool. If innovations arrive with an exogenous intensity, then a higher number of trades in a given interval must correspond to more liquidity trading. This is because informed traders in these markets have little incentive to split their orders – gas fees are levied per transaction and liquidity suppliers are not strategic and they do not need to hide their orders. Second, we look at volume assuming again that any volume beyond the informed volume must be noise. Finally we examine reversals, which we define as a trade that is immediately followed by a trade in the opposite direction of at least 75% of the size of the original trade size. This pattern should be observed whenever a liquidity trader who pushes the price away from its fundamental value is followed by an arbitrageur who brings the pool price back to the appropriate relative value. We cluster standard errors by pool and by day and run regressions with and without fixed effects.

Table 2 presents our results. Consistent with Hypothesis 1, we find that pool size decreases in token price volatility, which is our empirical proxy of the size of the innovation σ . Higher token price volatility means that liquidity suppliers in the pool lose more when picked off, thus liquidity providers in equilibrium are compensated with higher per token fee revenue – which is achieved by reducing the pool size. All three measures of uninformed trading are positively related to poolsize as predicted by our model. Our findings hold without fixed effects (columns (1)-(3)) as well as with pool and day fixed effects (columns (4)-(6)).

⁶Between September 18, 2021 and November 17, 2021 13.5 free UNI tokens per pool per block were distributed to users in four pools: WETH/USDT, WETH/USDC, WETH/DAI, and WETH/WBTC.

⁷Crypto volatilities are generally high and fluctuating, and our short term measure captures higher frequency market conditions consistent with many microstructure studies. Our findings are robust with respect to a 20 day moving average volatility.

	(1)	(2)	(3)	(4)	(5)	(6)
Std.Dev Fx Rate	-0.201*** (0.0419)	-0.249*** (0.0499)	-0.204*** (0.0428)	-0.163*** (0.0473)	-0.158*** (0.0487)	-0.152*** (0.0484)
Airdrop	222.9*** (26.93)	215.0*** (34.10)	214.2*** (35.00)	188.1*** (23.19)	186.2*** (25.31)	185.6*** (25.51)
Volume (USD)	0.741*** (0.198)			0.419*** (0.147)		
Reversals		49.18*** (13.68)			18.93*** (6.135)	
Number trades			6.633*** (2.088)			2.491*** (0.850)
R ²	0.263	0.215	0.213	0.552	0.529	0.529
Observations	997,507	997,507	997,507	997,507	997,507	997,507

Table 2. Regression explaining pool size as a function of price volatility and measures of uninformed trading. *Pool size* is the daily average size of the liquidity pool measured in million USD. *Airdrop* is a dummy set to one if transactions in that pool were rewarded with UNI tokens at that time, *Std.Dev FX Rate* is the annualized daily standard deviation of block by block price changes in the pool in percent. *Volume (USD)* is the daily trading volume in million USD, *Number Trades* is the number of trades for a pool per day in Thousands, and *Reversals* are defined as a trades that are immediately followed by an opposite trade of at least 75% of the size of the original trade measured in thousands. Standard errors are clustered by pool and by day. The last three columns include pool and day fixed effects. One, two, and three stars indicate significance at the 10%, 5%, and 1% level, respectively.

4.2 Stability in Liquidity provision

We focus on two important characteristics of liquidity supply. First, liquidity should be predictable for trades during the course of an average trading day. This reduces uncertainty and risk around the terms of trade. Second, during times of crisis, markets should still be viable, i.e., liquidity should be posted.

The strategic incentives of liquidity suppliers in limit order markets can lead to volatility in both cases. One of the characteristics of modern limit order markets is the rapid posting and cancelling of liquidity, and quote flickering. This is usually attributed to strategic liquidity provision. (Baruch and Glosten (2013) characterize equilibrium quote

flickering in a limit order book.) Further, “flash crashes” or rapid price movements caused by extreme liquidity supply withdrawals have been periodically observed in limit order markets.⁸ In this section, we investigate if AMM liquidity suppliers are similarly strategic, i.e., if they submit and withdraw liquidity at high frequency. We also examine stability in liquidity during crisis events. Our analysis in Section 5.2 below documents variation of price impact on a regular trading day, i.e., the effect of variation of liquidity supply on the realized terms of trade.

In principle, liquidity provision in an AMM could be as strategic as in a limit order market. To investigate if liquidity providers engage in high frequency strategic liquidity provision we track liquidity providers by the wallet address where they store their liquidity tokens. Out of our 94.2 million observations we find only 33,256 events where the same person deposited (withdrew) liquidity and withdrew (deposited) liquidity in the same pool within 50 blocks.⁹ Most of these withdrawals or additions are small, the median size is USD 1,797 or 2.76% of the pool, which has a very small impact on trading costs.¹⁰ We conclude that liquidity providers on Uniswap do not engage in high frequency strategic liquidity provision and withdrawals.

1.02% of all the interactions with Uniswap liquidity pools in our sample are withdrawals. Further, these withdrawals are not coincident with extreme market events. For example, on May 19, 2021 ETH dropped from over USD 3,400 to 2,014, a 41% decline. To put this in perspective, on October 19, 1987 the S&P 500 index only dropped by 20.5%. The orange area in Figure 9 shows minute by minute pricing data from Binance. We examine liquidity withdrawals from the large USD stablecoin pools that trade against

⁸The Securities and Exchange Commission (2010) provide a detailed analysis of the 2010 Flash Crash and document how HFT liquidity was withdrawn from the market. In addition, Kirilenko, Kyle, Samadi, and Tuzun (2017) analyze the crash.

⁹Only 387 observations are in pools with over USD 10,000 and with more than 1,000 transactions, where the deposit or withdrawal is for more than USD 1,000, and where the gap between deposit and withdrawal is no more than 5 blocks.

¹⁰Some large short term liquidity injections are due to scams where someone tries to inflate trading volume of a pool (e.g. 0x6bfd4393fcb5c2b803a2627bb996d66bdafe8859938145803a0c48739928d7d1c) or due to stabilization attempts of the FEI stable coin which used Uniswap as primary stabilization mechanism (e.g. 08883691096dd8624ab433124155536812a74585999bbf55958e92a6831d36df).

ETH: USDT, USDC, and DAI. The blue line in Panel A shows aggregate withdrawals from these pools in percent of pre-event poolsize. When Ether reached its lowest price, only about 2% of liquidity was withdrawn and users could trade with minimal price impact. Most withdrawals of liquidity happen around 53 minutes or 245 blocks after the big decline when prices have already recovered. Overall, only 17% of the liquidity gets withdrawn in this extreme price movement.

Two factors mitigate the liquidity providers' incentives to withdraw. First, increasing fee revenue partially offsets any losses liquidity providers experience. The green line in Panel A shows cumulative fee revenue during the plot period. Between 7am and 3pm UTC liquidity providers collected almost 3 million USD in fees. Trading volume and thus fee revenue might be high in periods of strong price movements because levered traders unwind their positions or are involuntarily liquidated (Lehar and Parlour (2023)) or because arbitrageurs take advantage of pricing differences across pools. Second, when liquidity providers want to withdraw liquidity they have to pay gas fees on the blockchain to compensate the miner for the computational cost of processing the transactions. Withdrawing liquidity requires the liquidity provider's share of the fee revenue to be computed which renders withdrawals computationally complex and thus more expensive relative to a swap. Panel B shows the median gas cost of a liquidity withdrawal. Gas costs spike as ETH drops making it uneconomical for many liquidity providers to withdraw.

Our model is single period and does not speak to pool dynamics. However from Figure 8 we can see that the optimal pool size decreases in the innovation of price σ , thus liquidity gets withdrawn when prices move a lot. While not explicitly modelled it is clear that high withdrawal costs would mitigate this effect. From the discussion in Section 4.1 we know that pool size increases in fee revenue. Higher fee revenue is therefore also mitigating the effect of higher volatility.

To test more broadly if high fee revenue and high gas fees coincide with periods of high volatility we collect daily token returns and fee revenue for all liquidity pools with at least

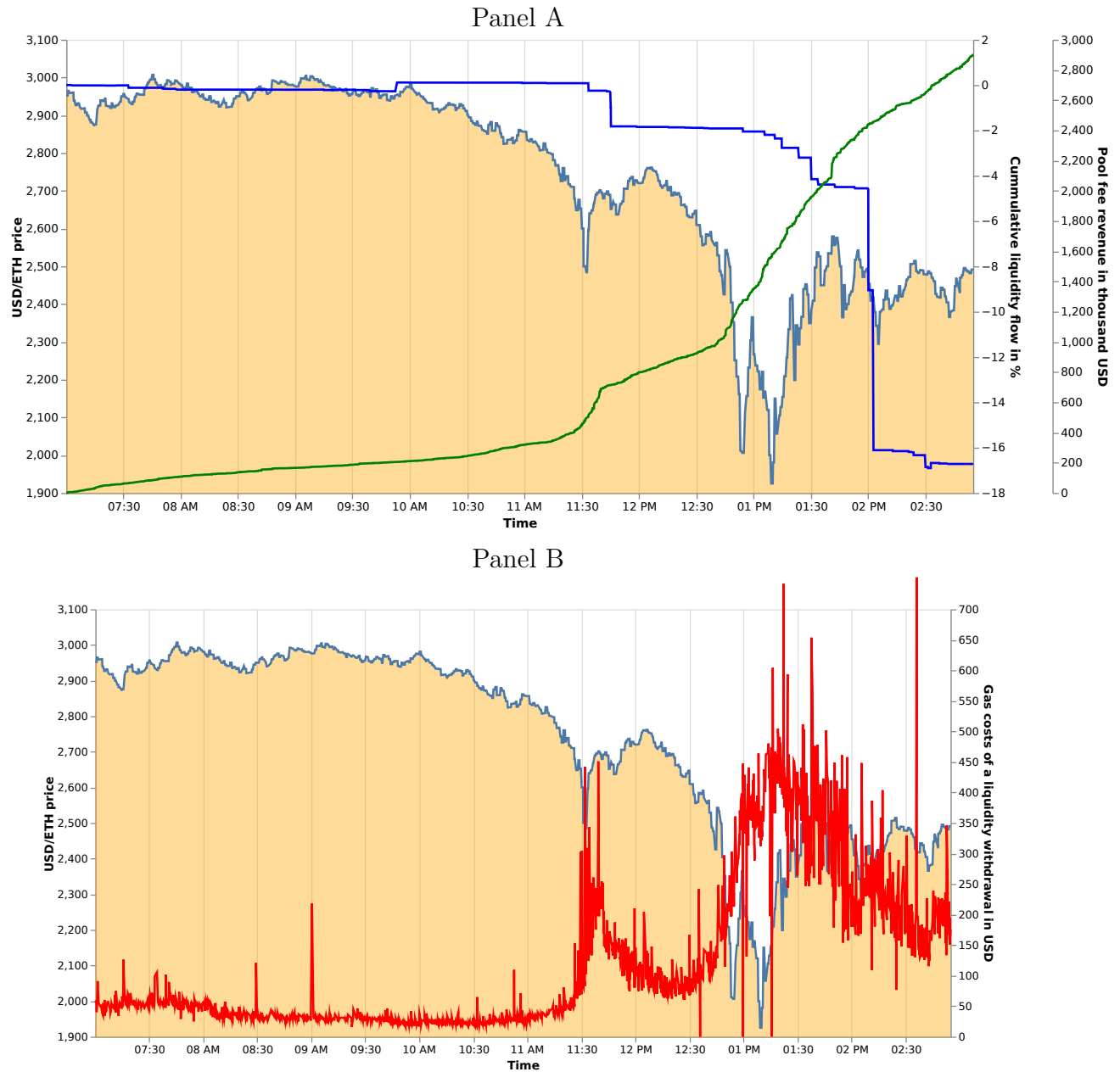


Figure 9. Liquidity withdrawals and extreme market events. Panel A shows the block by block price of Ether around May 19, 2020 (in orange, left axis), the aggregate percentage liquidity withdrawals from the four largest ETH - USD stablecoin pools on Uniswap (blue, right axis), and the fee revenue for the liquidity pools in thousand USD (green, far right axis). Panel B depicts the price of Ether and the median gas cost of a liquidity withdrawal in USD.

USD 10,000 in inventory and which are active for at least 30 days. We also collect average gas prices for all transactions that interact with a uniswap liquidity pool. Appendix F details our gas cost estimates. Variables are winsorized at the 95% level. We regress gas fees and fee revenue on the absolute return of the exchange rate in the pool as well as on a dummy set equal to one if the absolute daily return is more than 10% to measure extreme price movements. To interpret our regressions note that times of high price movements are necessarily times of high adverse selection costs, as these are the times that arbitrageurs trade. Table 3 presents our findings. We find that on days of extreme market movements gas costs of liquidity withdrawals are about USD 1.71 or 10% more expensive and pools earn USD 17.7 more in fees. Both gas costs and fee revenue increase in the absolute return.

	(1)	(2)	(3)	(4)
	Gas Fees Withdrawal (USD)		Fee revenue (USD)	
High return	1.705*** (0.0711)		17.73*** (2.575)	
Absolute return		17.26*** (0.593)		423.9*** (32.15)
Constant	16.41*** (0.00514)	15.89*** (0.0222)	257.0*** (0.186)	242.5*** (1.204)
R ²	0.155	0.155	0.325	0.326
Observations	1,303,869	1,303,869	1,303,869	1,303,869

Table 3. Regression explaining gas fees and fee revenue as a function of price variation. *Gas Fees Withdrawal* is the average daily gas cost of a liquidity withdrawal for a uniswap pool in USD, *Fee revenue (USD)* is the daily fee revenue of a liquidity pool, *High Return* is a dummy set to one if the token price in the pool changes by more than 10% on a day, *Absolute return* is absolute value of the daily change in the token price. Regressions include pool fixed effects and standard errors are clustered by pool. One, two, and three stars indicate significance at the 10%, 5%, and 1% level, respectively.

Rapid withdrawals of liquidity supply as we highlighted in our model are a feature of modern limit order markets. Further, such withdrawals have contributed to “flash crashes.” By contrast, liquidity suppliers in Uniswap pools are very stable. In as much as deep and

constant liquidity is socially beneficial, the design of the AMM is effective. We reiterate that Ethereum gas fees act as a commitment device for liquidity suppliers to remain in pools.

5 Trading costs comparison between a limit order market and an AMM

5.1 Theory

To compare the limit order market with an automated market maker we focus on the total trading cost for a liquidity trader, which consist of fees and price impact.¹¹ Recall, each liquidity supplier has undertaken private investment and is a monopolist with probability $\gamma^*(1 - \gamma^*)$. The liquidity trader thus faces a monopolist with probability $2\gamma^*(1 - \gamma^*)$, which we denote by η and enjoys competitive liquidity provision with probability $1 - \eta$.

Proposition 8 *On a limit order market, the expected cost to the liquidity trader per unit is*

$$E(c^{limit}) = \sigma((1 - \eta)\alpha + \eta).$$

On an active AMM pool, the expected cost to the liquidity trader per unit is

$$E(c^{AMM}) = p_0 \left(\frac{(1 + \tau)\lambda^b - (1 - \tau)\lambda^s}{2} \right),$$

where $\lambda^b(\tau, \alpha, \sigma) > 1$ and $\lambda^s(\tau, \alpha, \sigma) < 1$ are constants.

Trading costs on the limit order market are higher if the liquidity supplier is a monopolist,

¹¹Thus the objective function of the liquidity trader consists of a large private value for the trade minus trading costs.

and if there is more informed trading in the market. As noted, the payoffs to liquidity suppliers on the DEX are shared pro rata and so the equilibrium liquidity supply and hence cost to the liquidity trader are determined by the tradeoff between picking off risk and fees. These differences mean that the limit order market does not always dominate the automated market maker. When informed trading is low, the liquidity pool is large, price impact is low and thus cheaper for the liquidity trader. The price impact in the limit order market does not decrease in informed trading to the same extent because liquidity provision is not as competitive.¹²

Our model also implies that price impact in limit order markets is more volatile because the trader does not know ex-ante if he will face a monopolist liquidity provider or a competitive market. Since monopolist prices differ substantially from competitive ones we expect price impact to vary. For automated market makers price impact is known ex-ante and can be very low when the pool is sufficiently large.

Proposition 9 *The limit order market does not dominate the automated market maker.*

- i. If $\frac{\tau}{\eta} < \frac{\sigma}{p_0}$ there exists a critical innovation probability α^* such that the liquidity trader strictly prefers the AMM whenever $\alpha < \alpha^*$.*
- ii. Conditional on trading quantity q price impact is more volatile in the limit order market than the AMM.*

Figure 10 depicts the region of the parameter space where total trading costs for the liquidity trader in the liquidity pool are lower than in the limit order market. The left panel shows that automated market makers are the better trading venue for the liquidity trader when either volatility, σ , or the probability of an innovation and attendant informed trading, α , are sufficiently small. In those cases the pool is large and the liquidity trader can trade without much price impact. The right panel shows that for a given probability

¹²Market makers invest heavily in high frequency trading which is wasteful investment for the liquidity trader and deters entry in liquidity provision.

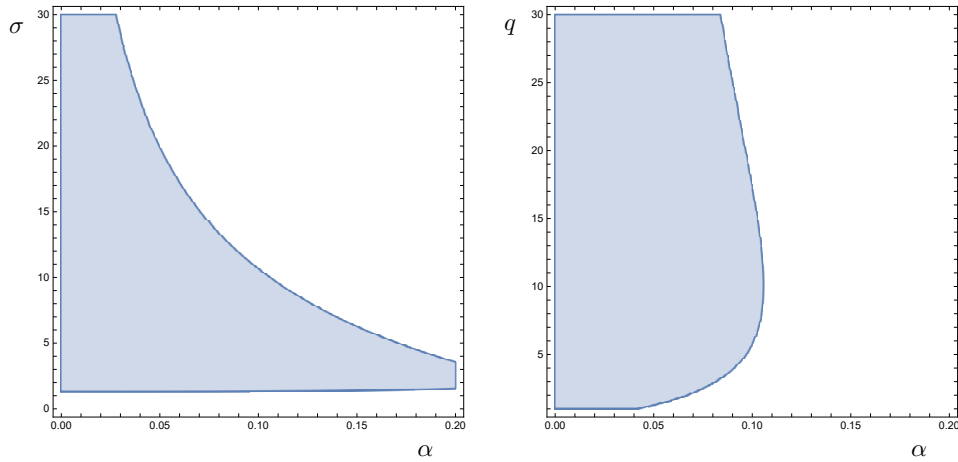


Figure 10. Region of the parameter space where AMMs dominate limit order markets. The graph shows the region of the parameter space where trading costs for the liquidity trader in the liquidity pool are lower than in the limit order market. Unless otherwise stated the parameters are $P_0 = 100, \sigma = 10, q = 10, a = 45, \tau = 0.003$

of an innovation liquidity pools dominate when the trade size of the liquidity trader is neither too small, which generates a low fee revenue, nor too large.

Our definition of AMM transaction costs includes the fee τ while our definition of transaction costs on the limit order market does not include any fees (make/take fees). Also notice that the trading cost on the AMM differs from price impact. This is because on the AMM each unit trades at a different marginal price. The equivalent notion on a limit order book would be the total amount paid as an order walked the book. Empirically, we observe the price after each transaction (the marginal price) and not the total amount paid.

There are three additional reasons why an comprehensive empirical analysis of the differences in costs is challenging. First, as we discuss in Section 5.2 below, assets traded on an AMM differ from those traded on a centralized exchange. In addition to the non-fungibility of assets, trader sorting across the two venues is likely. Second, to estimate

price impact on a limit order market, quote data are important.¹³ This is because aggressive orders always cross the spread and pay it. Finally, order submission strategies on both venues differ, as there is an incentive to split orders on limit markets. Based on Observation 9 and with these caveats in mind, we formulate the following empirical hypotheses:

Hypothesis 2 *We expect AMMs to have lower price impact relative to the limit order market when volatility is low, there is more noise trading, and for medium trade sizes.*

Hypothesis 3 *We expect AMMs to have lower variation in price impact relative to the limit order market.*

5.2 Empirical Analysis

Our framework does not incorporate market access fees. However, various papers such as Capponi and Jia (2021) and Barbon and Rinaldo (2021) emphasize gas fees as an important determinant of Uniswap market quality. Fully quantifying the difference between on-chain markets (such as Uniswap) and off-chain markets (such as Binance) is complex. First, a tokens on-chain and off-chain differ in custody risk and use value. Second, off-chain venues have their own market access fees and in some case make-take fees.

Binance, as all ‘centralized’ crypto exchanges retains custody of traded assets. As the recent collapse of FTX illustrated, this custody risk is not insignificant. Uniswap, in contrast, is non-custodial, meaning that the user can initiate a trade directly out of their personal wallet and keep custody of traded assets until they are swapped in an atomic transaction.

Uniswap traded tokens can also be used in other DeFi protocols such as for yield farming or as collateral. Binance based tokens have to be transferred out of Binance custody into

¹³With richer data on Binance but a smaller sample Barbon and Rinaldo (2021) confirm our findings.

a user's wallet before they can be used in other protocols. Similarly, any tokens that a user wants to trade on the exchange must be transferred out the user's personal wallet into the exchange wallet and the exchange has to give the user credit for these assets in their own internal ledger before they can be traded. Both steps incur transaction costs.

In short, owning a token on Binance has different attributes to owning a token on chain. The difference in value should account for the opportunity cost of being off-chain and having access to the different protocols, and the direct cost of being exposed to Binance credit risk.

We can quantify blockchain transaction costs by comparing a swap transaction on a Dex to two tokens transfers, which would be required to move a token in and out of a centralized exchange. Gas cost are the product of gas amount and gas price. The former is determined by the computational complexity of the transaction and the latter is set by the user. We collect blockchain data for both, and present the details of the estimate in Appendix F. The average incremental fee of trading on a Dex is 0.94 cents, the median incremental fee is 0.22 cents. On some days the incremental fee can be very high. Indeed, the maximum in our sample is USD 11.57 on May 11, 2021 when ETH first broke through USD 4,000. For the latter part of our sample, since January 2021, the median incremental fee was USD 1.33. Finally, observe that our calculation does not include any withdrawal fees that an exchange may charge. Binance, for example, charges token-specific fees that are hard to include in a comparison. Binance charges a fee of 0.005 ETH for ETH withdrawals to the Ethereum blockchain, which at the time of writing the paper would be equivalent to USD 8.50.

To compare Uniswap to traditional exchanges we collect minute by minute trading data from Binance (international), one of the largest crypto-exchanges by volume. Many of the 2,148 token pairs listed on Binance trade against fiat currencies. We find 166 token pairs that trade on both Uniswap and Binance, however many of these pairs are very infrequently traded. We eliminate all pairs with an average daily volume of less than 10

ETH on either market, less than 90 trading days, and an average daily Uniswap pool size of less than 10 ETH. We also eliminate all observations in the first 90 days of the initiation of trading on either venue and where the pricing error between the two venues is more than 10%. We treat WETH and ETH as identical given the easy and cheap conversion. We end up with 43 token pairs that are cross listed on Uniswap and Binance.

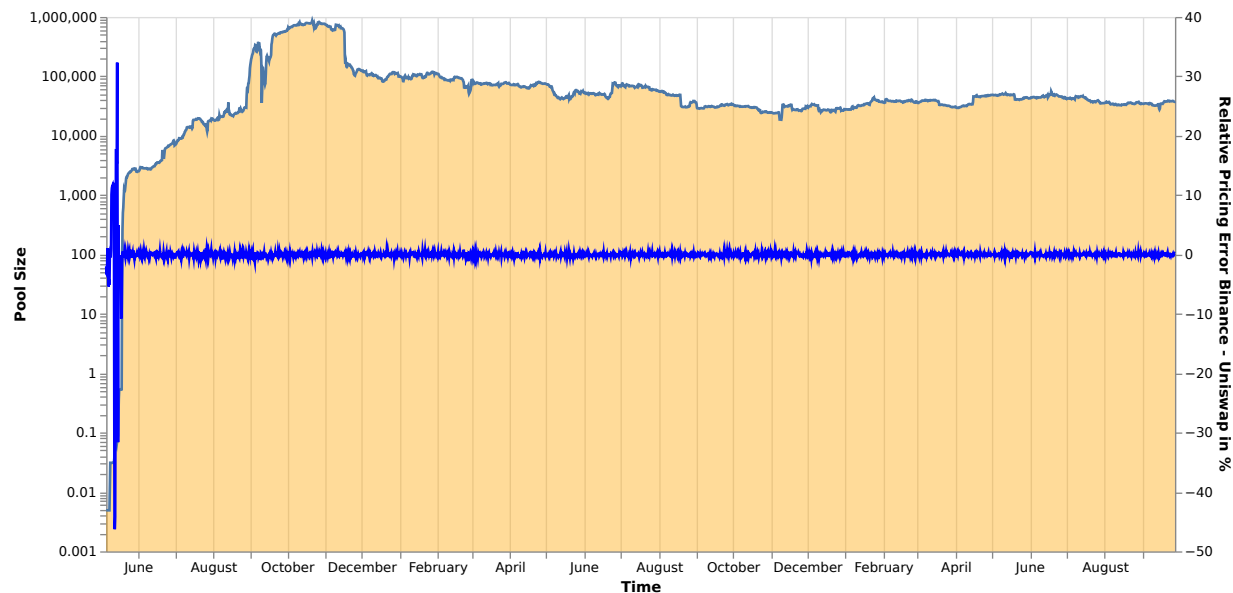


Figure 11. Pricing error and pool size Pricing error for the USDC/ETH pair when comparing Binance to Uniswap in percent of the Binance price (blue line, right axis) and pool size of the Uniswap USDC/ETH pool (orange, log-scale, left axis). Prices are calculated at minute intervals. The graph plots these errors sampled every eight hours.

Pricing differences between Uniswap and Binance are small except in the startup-phase of the Uniswap pool when liquidity is scarce. Figure 11 shows the pricing in blue and pool size (in orange, log scale) for the USDC/ETH pool. We calculate these by taking the closing price per minute on both Binance and Uniswap. The pricing error is calculated as

$$\frac{\text{Price on Binance} - \text{Price on Uniswap}}{\text{Price on Binance}}$$

When a pool starts, as long as the poolsize is below 100 ETH, pricing errors are huge reaching over 40%. This is not surprising as a small invariant k will cause a very steep

bonding curve (see equation 1) and make arbitrage unprofitable. Once the poolsize is above 700 ETH, the pricing difference stays below 1.3% with an average absolute pricing error of 0.19% for this pool.



Figure 12. Intraday prices for the USDC/ETH pair on October 21, 2020 The graph shows minute-by-minute prices of the USDC/ETH pair on Binance and Uniswap.

Figure 12 shows intraday prices of the USDC/ETH pair on one day, October 21, 2020. The pattern is typical for most days in our sample. Binance prices often lead Uniswap prices and as expected Binance prices are more volatile than those on Uniswap.

Following Proposition 9, we expect some Uniswap pools to have lower price impact relative to a limit order market such as Binance. We define price impact as the absolute change in price over change in quantity traded in a one minute interval. We then average price impact measures over a day. Figure 13 shows our measures for price impact for the USDC/ETH pair. We can see that price impact on Binance almost always exceeds that on Uniswap. Binance's price impact also varies over time while the price impact on Uniswap stays constant and is negligible once the pool has reached an equilibrium size (in September 2020). We also compute the theoretical price impact for Uniswap which we

derive analytically from the bonding curve in Equation (1) assuming zero fees. We find that our analytical measure of price impact on Uniswap (green line) corresponds closely to our empirical measure (orange line).

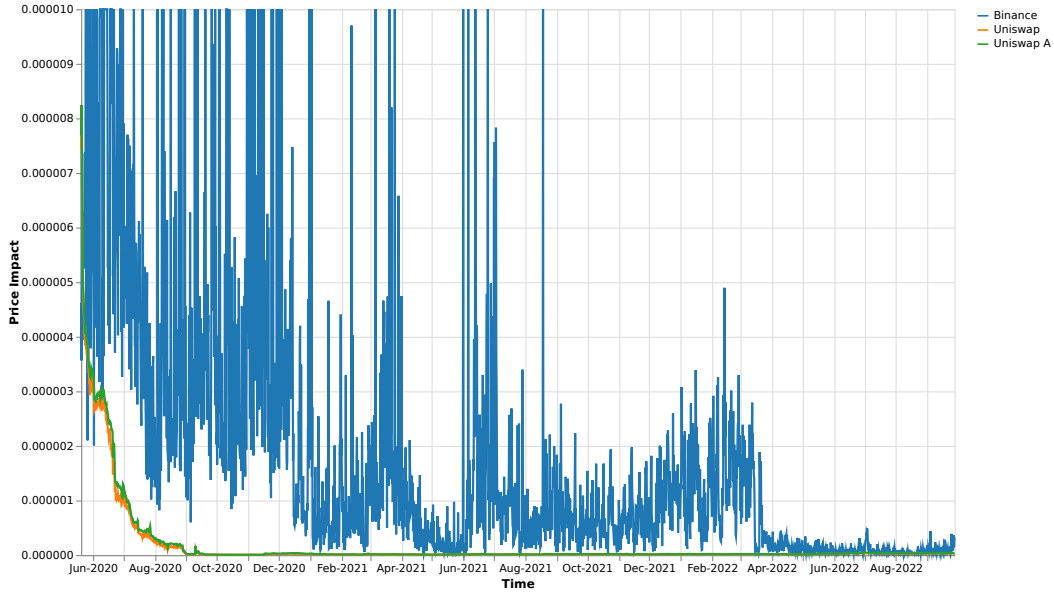


Figure 13. Price Impact of USDC/ETH on Uniswap (orange, green) and Binance (blue). Price impact is computed as change in price over volume (green and blue lines) as well as analytically as the price change for a marginal unit bought using the bonding curve formula (green line). Binance price impact is cut off at 0.00001.

We observe a large drop in the price impact on Binance as of March 2022. On March 25, Binance integrated Pancake swap into their trading app. Pancake swap is a Dex similar to Uniswap running on the Binance Smart Chain (BSC). Simultaneously Binance introduced the Binance DeFi wallet which allows users of the centralized exchange to open a wallet and use DeFi application on BSC out of the app (see e.g. Binance Blog (2022)). The integration of a Dex into the Binance trading app is associated with a substantial drop in price impact on Binance.

To analyze price impact more formally we define relative price impact rPI as

$$rPI = \frac{PI_{Binance}}{PI_{Uniswap}} - 1 \quad (18)$$

where $PI_{Binance}$ and $PI_{Uniswap}$ are the daily price impact measures for a token pair on Binance and Uniswap, respectively. We compute the daily price impact as the median price impact over all one-minute intervals in a day where there was positive volume.¹⁴ This relative measure controls for token and day specific factors that influence price impact in both markets. To test Hypothesis 2 we run the following regression:

$$rPI = a + b_1 \mathbb{1}_{pancake} + b_2 \sigma_{FX} + b_3 trade\ size + b_4 trade\ size^2 + b_5 noise\ trading + \epsilon \quad (19)$$

To control for the Pancake Swap integration into Binance we set the dummy $\mathbb{1}_{pancake}$ to one after March 25, 2022. We expect Binance price impact to align closer with that of Uniswap after the integration of a Uniswap-clone into Binance. Trade size is the combined volume of Binance and Uniswap in ETH divided by the combined number of trades on Binance and Uniswap. As in the numerical example in Figure 10 we expect AMMs to have lower price impact for middle trade sizes, and thus rPI to be u-Shaped in trade size. Our measure of noise trading is the daily volume of trades that are immediately reversed on Uniswap measured in ETH. Standard errors are clustered by pool and by day. Models that include the pancake dummy only have pool fixed effects. For robustness we run the same regression with pool and day fixed effects without the dummy.

Table 4 presents our findings. We first note that the intercept is positive and significant, demonstrating that price impact on Binance is on average higher than on Uniswap. The Dex integration on Binance as measured by the Pancake dummy is associated with a significant drop in Binance price impact relative to Uniswap. Consistent with this Hypothesis 2 we find that the price impact on Binance decreases in volatility relative to Uniswap. As predicted our measure of relative price impact is first increasing, then decreasing in trade size. Higher reversal volume, which proxies for less informed trading, is associated with higher relative price impact on Binance.

¹⁴We measure absolute price change over volume which corresponds to a high frequency version of the Amihud (2002) measure.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Pancake Swap	-3.013*** (0.728)	-3.085*** (0.728)	-2.405*** (0.657)	-1.931*** (0.599)	-1.941*** (0.590)				
Std.Dev Fx Rate		-2.902*** (0.707)			-3.034*** (0.662)	-2.418*** (0.754)	-2.497*** (0.735)	-2.963*** (0.979)	-0.381 (0.332)
Trade Size			1.472*** (0.236)		1.118*** (0.317)		0.939*** (0.297)	1.000*** (0.261)	1.149 (1.045)
Trade Size ²			-0.0120*** (0.00290)		-0.00900*** (0.00258)		-0.00742*** (0.00256)	-0.00791*** (0.00257)	-0.0759 (0.0598)
Reversal Volume				2.731*** (0.712)	1.583** (0.745)	1.776** (0.726)	0.910 (0.725)		
Constant	4.080*** (0.286)	4.956*** (0.387)	3.126*** (0.303)	2.644*** (0.350)	3.394*** (0.350)	3.015*** (0.336)	2.900*** (0.258)	3.963*** (0.302)	1.559*** (0.179)
R ²	0.446	0.457	0.545	0.520	0.576	0.610	0.636	0.651	0.826
Observations	24,963	24,224	24,963	24,963	24,224	24,216	24,216	15,387	8,829
Fixed effects	Pool	Pool	Pool	Pool	Pool	Pool,Day	Pool,Day	Pool,Day	Pool,Day

Table 4: **Regression explaining relative price impact between Uniswap and Binance markets.** *Relative price impact* is the Binance price impact divided by the Uniswap price impact minus one, *Pancake Swap* is a dummy set to one after March 25, 2022, the day when Binance integrated Pancake Swap, a DEX, into the Binance app, *Std.Dev FX Rate* is the annualized daily standard deviation of block by block price changes on Uniswap, *Trade Size* is the average trade size computed as the aggregate volume of Binance and Uniswap in ETH divided by the number of trades on Binance and Uniswap, *Trade Size*² is the squared trade size, *Reversal Volume* is the volume in ETH of trades on Uniswap that are reversed immediately, a proxy for noise trading. Columns (1) - (7) cover the whole sample, while columns (8) and (9) present results the sub-samples before and after the pancake swap integration into Binance, respectively. Regressions include pool (i.e. token pair) fixed effects and day fixed effects in columns (6)-(9). Standard errors are clustered by pool and by day. One, two, and three stars indicate significance at the 10%, 5%, and 1% level, respectively.

The last two columns of Table 4 present results for the subsample before the Pancake swap integration in column (8) and after in column (9). We find that most variables become insignificant after the Pancake Swap integration. The intercept is smaller but still significant post integration documenting that Uniswap has lower price impact in both subsamples.

Hypothesis 3 predicts that price impact is more volatile on the limit order market. To test this prediction compute 30 day moving average volatility V of our daily price impact measure. We then define relative volatility as

$$rV = \frac{V_{Binance}}{V_{Uniswap}} - 1 \quad (20)$$

which again controls for unobserved token and day specific factors that influence volatility of price impact in both markets. We use the same empirical specification as in Equation (19). Table 5 presents our findings. From the positive intercept we see again that price impact is more volatile on Binance and drops significantly after the Pancake Swap integration. Similar to price impact we document that price impact on Binance is more volatile when the exchange rate volatility is low, for medium trade sizes, and when there is more informed trading (lower reversal volume).

The relative attractiveness of AMMs attracts traders to decentralized exchanges. Figure 14 shows the trading volume of the USDC/ETH pair on Binance and Uniswap, respectively. We can see that trading volume is remarkably correlated across the two markets, which is surprising given that tokens have to be moved back and forth through on-chain transactions between the two markets. We can also see that Uniswap is gaining market share over time and eventually more trading is happening on Uniswap relative to Binance. Towards the end of the sample trading volume on Uniswap V2 declines, most likely due to the fact that volume moves the Uniswap V3.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Pancake Swap	-13.63** (5.592)	-13.79** (5.437)	-10.18* (5.079)	-7.916 (4.821)	-7.907* (4.613)				
Std.Dev Fx Rate		-21.48*** (4.869)			-22.23*** (4.653)	-16.43*** (4.317)	-17.37*** (4.294)	-18.16*** (5.829)	-7.756*** (2.551)
Trade Size			7.202*** (1.359)		5.826*** (1.151)		4.991*** (1.177)	5.272*** (1.266)	20.16*** (7.192)
Trade Size ²			-0.0556*** (0.0193)		-0.0439*** (0.0144)		-0.0386*** (0.0132)	-0.0409*** (0.0149)	-1.430*** (0.446)
Reversal Volume				11.73** (5.038)	6.228 (3.859)	9.719* (5.046)	5.421 (4.016)		
Constant	33.95*** (1.992)	39.00*** (2.512)	29.01*** (1.857)	26.99*** (2.752)	31.44*** (2.597)	28.99*** (2.339)	28.44*** (1.863)	30.47*** (1.539)	25.91*** (1.216)
R ²	0.416	0.427	0.457	0.441	0.474	0.517	0.530	0.523	0.677
Observations	21,409	21,275	21,409	21,409	21,275	21,266	21,266	13,981	7,284
Fixed effects	Pool	Pool	Pool	Pool	Pool	Pool,Day	Pool,Day	Pool,Day	Pool,Day

Table 5: **Regression explaining relative volatility of price impact between Uniswap and Binance markets.** *Relative volatility of price impact* is the volatility of Binance price impact divided by the volatility of uniswap price impact minus one, *Pancake Swap* is a dummy set to one after March 25, 2022, the day when Binance integrated Pancake Swap, a DEX, into the Binance app, *Std.Dev Fx Rate* is the annualized daily standard deviation of block by block price changes on Uniswap, *Trade Size* is the average trade size computed as the aggregate volume of Binance and Uniswap in ETH divided by the number of trades on Binance and Uniswap, *Trade Size*² is the squared trade size, *Reversal Volume* is the volume in Eth of trades on Uniswap that are reversed immediately, a proxy for noise trading. Columns (1) - (7) cover the whole sample, while columns (8) and (9) present results the sub-samples before and after the pancake swap integration into Binance, respectively. Regressions include pool (i.e. token pair) fixed effects and day fixed effects in columns (6)-(9). Standard errors are clustered by pool and by day. One, two, and three stars indicate significance at the 10%, 5%, and 1% level, respectively.

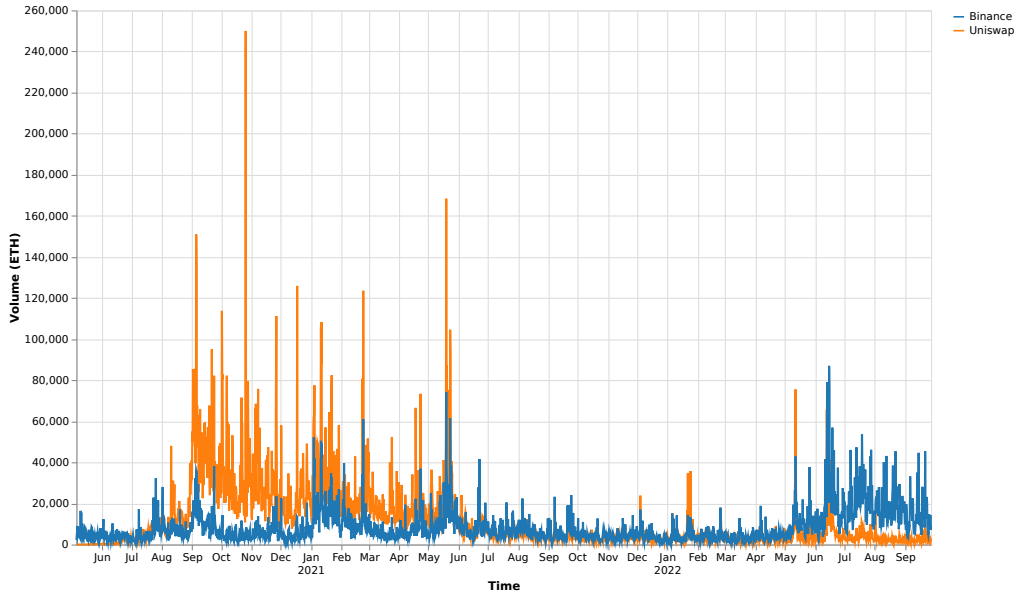


Figure 14. Trading volume of USDC/ETH on Uniswap (orange) and Binance (blue). The graph shows the trading volume excluding flash loans in ETH. Trading volume is aggregated over rolling eight hour intervals.

6 Conclusion

In 1971, Fischer Black wrote two articles for the *Financial Analyst's Journal* speculating on whether computers or “automation” could ever replace human interaction in financial markets (Black (1971a), Black (1971b)). In these papers, he argued that market liquidity was constrained by the size of a market maker’s inventory and suggested that a solution would be to have more direct participation from other market participants.

The Uniswap experiment does not merely increase the supply of liquidity by relaxing market makers’ inventory constraints. It also changes how the benefits and costs of liquidity provision are shared among market participants. In a limit order market, the exchange sets price and time priority, which effectively determine the interaction of liquidity demand and supply. Price impact in a limit order market is the endogenous outcome of the interaction. By contrast, on an AMM the price impact is programmatically determined by a bonding curve. However, we note that this impact is conditional on the size of the

pool. Thus, the equilibrium price impact arises as the pool size adjusts, so that liquidity suppliers trade off potential adverse selection against fee revenue.

The automated Uniswap protocol has clearly been successful. Further, in July 2021, Swarm a Berlin based DeFi company announced that it was launching an AMM that was fully licensed by the German regulator, BaFin. One of the reasons for the uptake of these liquidity sharing protocols is that we have demonstrated both theoretically and empirically that pools adjust to tradeoff the benefits and costs of liquidity provision. Further, compared with a centralized exchange there are some token pairs for which the AMM provides liquidity more efficiently than a centralized exchange.

References

- Amihud, Yakov, 2002, Illiquidity and stock returns: cross-section and time-series effects, *Journal of Financial Markets* 5, 31–56.
- Angeris, Guillermo, and Tarun Chitra, 2020, Improved price oracles: Constant function market makers, in *Proceedings of the 2nd ACM Conference on Advances in Financial Technologies* pp. 80–91.
- Angeris, Guillermo, Hsien-Tang Kao, Rei Chiang, Charlie Noyes, and Tarun Chitra, 2021, An Analysis of Uniswap markets, *Cryptoeconomic Systems* 0.
- Aoyagi, Jun, 2020, Liquidity Provision by Automated Market Makers, working paper, HKUST.
- Barbon, Andreas, and Angelo Ranaldo, 2021, On The quality of Cryptocurrency Markets, working paper, St. Gallen University.
- Baruch, Shmuel, and Lawrence R. Glosten, 2013, Fleeting Orders, working paper.
- Biais, Bruno, Thierry Foucault, and Sophie Moinas, 2015, Equilibrium Fast Trading, *Journal of Financial Economics* 116, 292–313.
- Binance Blog, 2022, First DeFi Project on Binance: The PancakeSwap Mini Program Bringing the CeFi and DeFi Ecosystem Together, <https://www.binance.com/en/blog/markets/first-defi-project-on-binance-the-pancakeswap-mini-program-bringing-the-cefi-and-defi>
- Black, Fisher, 1971a, Toward a Fully Automated Stock Exchange-Part 1, *Financial Analysts Journal* 27, 28–35.
- Black, Fisher, 1971b, Toward a Fully Automated Stock Exchange, Part 2, *Financial Analysts Journal* pp. 86–87.
- Brogaard, Jonathan, Terrence Hendershott, and Ryan Riordan, 2019, Price Discovery without Trading: Evidence from Limit Orders, *Journal of Finance* 74, 1621–1161.
- Capponi, Agostino, and Ruizhe Jia, 2021, The Adoption of Blockchain-based Decentralized Exchanges, working paper, Columbia University.
- Glosten, Lawrence R., 1994, Is the Electronic Open Limit Order Book Inevitable?, *Journal of Finance* 49, 1127–1658.
- Glosten, Lawrence R., 1998, Precedence Rules, working paper.
- Hao, Jia, Avner Kalay, and Stewart Mayhew, 2009, Ex-Dividend Arbitrage in Option Markets, *The Review of Financial Studies* 23(1), 271–303.

- Haynes, Richard, and Esen Onur, 2020, Precedence Rules in Matching Algorithms, *Journal of Commodity Markets* 19.
- Ito, Yuki, and Jun Aoyagi, 2021, Co-Existing Exchange Platforms, working paper, HKUST.
- Kirilenko, Andrei, Albert S. Kyle, Mehrdad Samadi, and Tugkan Tuzun, 2017, The Flash Crash: High-Frequency Trading in an Electronic Market, *The Journal of Finance* 72, 967–998.
- Lehar, Alfred, and Christine Parlour, 2023, Systemic Fragility in Decentralized Markets, working paper, UC Berkeley.
- Park, Andreas, 2021, The Conceptual Flaws of Constant Product Automated Market Making, working paper, University of Toronto.
- Securities and Exchange Commission, 2010, Findings Regarding the Market Events of May 6, 2010, Working paper, SEC.

A Proofs

Proof of Lemma 1

If the liquidity supplier is alone in the market, and posts at prices $\underline{p}$ at which he buys and $\bar{p}$ at which he sells, with $\underline{p} < \bar{p}$ then

- i. With probability $(1 - \alpha)$ a noise trader arrives. If the noise trader is a buyer, the liquidity supplier sells to him at $\bar{p}$ and obtains a payoff of $(\bar{p} - p_0)$. Symmetrically, if the noise trader is a seller, the liquidity supplier buys from him at $\underline{p}$ and obtains a payoff of $(p_0 - \underline{p})$.
- ii. With probability α there is an information event. If the informed trader buys, the liquidity supplier obtains a payoff $\bar{p} - (p_0 + \sigma)$ and if the informed trader sells, he obtains a payoff of $(p_0 - \sigma) - \underline{p}$.

His overall profit is then

$$(1 - \alpha) \left[\frac{1}{2}(\bar{p} - p_0) + \frac{1}{2}(p_0 - \underline{p}) \right] + \alpha \left[\frac{1}{2}(\bar{p} - (p_0 + \sigma)) + \frac{1}{2}((p_0 - \sigma) - \underline{p}) \right] \quad (21)$$

Clearly, he will post a sell price at which to sell of $\bar{p} = p_0 + \sigma$, and a price at which to buy of $\underline{p} = p_0 - \sigma$, to obtain a profit of $(1 - \alpha)\sigma$.

■

Proof of Lemma 2

When two limit order submitters are in competition Nature will select one liquidity provider (LP) denoted by i to go first and he will post a price of p_i . If this price is below the maximum price of $p_0 + \sigma$ and leaves i with a profit then the second LP will undercut i by posting a price of $p_j = p_i - \varepsilon$. LP j will absorb an order for q either by the noise trader or by the arbitrageur. The noise trader's demand is satisfied, however the arbitrageur would also exhaust the liquidity of i causing a loss if $p_0 + \sigma - p_i$ for the LP. Therefore any price p_i that leaves LP i with a profit cannot be an equilibrium.

The only price for which j would not undercut is the price where i makes zero profit, which is defined by

$$\alpha(p_i - (p_0 + \sigma)) + (1 - \alpha)(p_i - p_0) = 0 \quad (22)$$

or $p_i = p_0 + \alpha\sigma$.

LP j cannot undercut i and knows that he will only potentially trade with the arbitrageur. The arbitrageur trades when the price is $p_0 + \sigma$ and to avoid any loss LP j sets $p_j = p_0 + \sigma$.

Both LPs make zero profit in expectation. ■

Proof of Proposition 3

The ex ante profit for the limit order trader i is

$$\gamma_i(1 - \gamma_j)(1 - \alpha)\sigma q - I(\gamma_i)$$

The reaction function is defined by

$$(1 - \gamma_j)(1 - \alpha)\sigma q - 2a\gamma_i = 0$$

In symmetric equilibrium, each liquidity supplier will choose a private level of investment

$$\gamma^* = \frac{(1 - \alpha)\sigma q}{2a + \sigma q(1 - \alpha)}$$

■

Proof of Lemma 4

The fees accruing to the liquidity providers from accommodating the liquidity trader follows from the text, and are τe_ℓ^s and τe_ℓ^b ,

where

$$\begin{aligned} e_\ell^b &= \frac{E_0 T_0}{T_0 - q} - E_0 \\ e_\ell^s &= E_0 - \frac{E_0 T_0}{T_0 + q}. \end{aligned}$$

The arbitrageur reverses the trade and pays the same fee in Eth. The overall expected fee revenue is then

$$\begin{aligned} &= 2\tau \left(\frac{1}{2} \frac{E_0 T_0}{T_0 - q} - \frac{1}{2} \frac{E_0 T_0}{T_0 + q} \right) \\ &= \tau E_0 T_0 \left(\frac{T_0 + q - (T_0 - q)}{(T_0 - q)(T_0 + q)} \right) \\ &= 2\tau p_0 T_0^2 \left(\frac{q}{T_0^2 - q^2} \right), \end{aligned}$$

where the last line uses the fact that $p_0 = \frac{E_0}{T_0}$.

Reversing the trade is profitable for the arbitrageur if the AMM price including transaction costs differs from the fundamental value or

$$\begin{aligned} e_\ell^b(1 - \tau) \geq p_0 & \quad \text{or} \quad \frac{T_0 q}{(T_0 - q)} > \frac{1}{1 - \tau} & \text{to reverse a liquidity buy order} \\ e_\ell^s(1 + \tau) \leq p_0 & \quad \text{or} \quad \frac{T_0 q}{(T_0 + q)} < \frac{1}{1 + \tau} & \text{to reverse a liquidity sell order,} \end{aligned}$$

which yields the conditions,

$$\begin{aligned} \tau < 1 - \frac{T_0 - q}{T_0 q} & \quad \text{To reverse the liquidity buy} \\ \tau < \frac{T_0 + q}{T_0 q} - 1 & \quad \text{To reverse the liquidity sell} \end{aligned}$$

Define $\tau^* = \min \left[1 - \frac{T_0 - q}{T_0 q}, \frac{T_0 + q}{T_0 q} - 1 \right]$. Then, for τ, τ^* , this trade is profitable for the arbitrageur. ■

Proof of Lemma 5

The change in value to the liquidity providers if there is a positive asset innovation follows directly from the arguments in the text.

If there is a negative asset innovation, the post innovation value of the committed capital is $E_0 + T_0(p_0 - \sigma) = 2E_0 - \sigma T_0$. If there is a negative asset innovation, then the informed trader sells up to the point of no arbitrage, and the trade occurs on the bonding curve, so

$$(E_0 - e_I^s)(T_0 + t^s) = E_0 T_0. \quad (23)$$

$$e_I^s = E_0 - \frac{E_0 T_0}{T_0 + t^s} \quad (24)$$

The profits of the informed trader are given by

$$\pi_I^s = (1 - \tau)e_I^s - (p_0 - \sigma)t^s \quad (25)$$

$$= (1 - \tau) \left(E_0 - \frac{E_0 T_0}{T_0 + t^s} \right) - (p_0 - \sigma)t^s. \quad (26)$$

The first order condition is

$$(1 - \tau) \frac{E_0 T_0}{(T_0 + t^s)^2} - (p_0 - \sigma) = 0. \quad (27)$$

Note, this represents optimal trading if $t^s = \sqrt{\frac{(1-\tau)E_0T_0}{p_0-\sigma}} - T_0 \geq 0$, or $\frac{\sigma}{p_0} \geq \tau$.

Thus, we obtain

$$\begin{aligned} t^s &= \sqrt{\frac{(1-\tau)E_0T_0}{p_0-\sigma}} - T_0 \\ e_I^s &= E_0 - \sqrt{\frac{E_0T_0(p_0-\sigma)}{1-\tau}}. \end{aligned}$$

Hence, the value of the committed capital after the asset value has dropped and the informed trader has traded is

$$E_0 - e_I^s + (T_0 + t^s)(p_0 - \sigma) = \left(\frac{2-\tau}{\sqrt{1-\tau}} \right) \sqrt{E_0T_0(p_0-\sigma)}.$$

Thus, the change in total value of the committed capital is

$$\left(\frac{2-\tau}{\sqrt{1-\tau}} \right) \sqrt{E_0T_0(p_0-\sigma)} - 2E_0 + \sigma T_0. \quad (28)$$

The liquidity cost paid by the informed trader is $\tau \left(E_0 - \sqrt{\frac{E_0T_0(p_0-\sigma)}{1-\tau}} \right)$, so the total payoff to liquidity provision if the asset value drops is

$$\left(\frac{2-\tau}{\sqrt{1-\tau}} \right) \sqrt{E_0T_0(p_0-\sigma)} - 2E_0 + \sigma T_0 + \tau \left(E_0 - \sqrt{\frac{E_0T_0(p_0-\sigma)}{1-\tau}} \right) \quad (29)$$

$$= \left(\frac{2-\tau}{\sqrt{1-\tau}} \right) \sqrt{T_0^2 p_0 (p_0 - \sigma)} - 2p_0 T_0 + \sigma T_0 + \tau \left(T_0 p_0 - \sqrt{\frac{T_0^2 p_0 (p_0 - \sigma)}{1-\tau}} \right) \quad (30)$$

$$= 2\sqrt{1-\tau} \sqrt{T_0^2 p_0 (p_0 - \sigma)} - (2-\tau)p_0 T_0 + \sigma T_0 \quad (31)$$

■

Proof of Proposition 6

From Lemmas 4 and 5, the overall payoff to liquidity provision (for the entire pool) and using the fact that $p_0 = \frac{E_0}{T_0}$ is

$$\begin{aligned}
& (1 - \alpha)2\tau p_0 T_0^2 \left(\frac{q}{T_0^2 - q^2} \right) \\
& + \frac{\alpha}{2} \left(2T_0 \sqrt{(1 + \tau)} \sqrt{p_0(p_0 + \sigma)} \right. \\
& - (2 + \tau)p_0 T_0 - \sigma T_0 \left. \right) + \frac{\alpha}{2} \left(2T_0 \sqrt{1 - \tau} \sqrt{p_0(p_0 - \sigma)} - (2 - \tau)p_0 T_0 + \sigma T_0 \right)
\end{aligned}$$

Or,

$$\frac{(1 - \alpha)2\tau p_0 T_0^2 q}{T_0^2 - q^2} + T_0 \alpha \left(\sqrt{p_0(p_0 + \sigma)(1 + \tau)} + \sqrt{(1 - \tau)p_0(p_0 - \sigma)} - 2p_0 \right) \quad (32)$$

The liquidity suppliers have deep pockets with a zero opportunity cost of capital, and are indifferent between committing extra value when the value to liquidity provision is zero. Thus, the equilibrium size of the liquidity pool is implicitly defined by setting Equation 32 equal to zero, which yields,

$$0 = (1 - \alpha)2\tau p_0 T_0 \left(\frac{q}{T_0^2 - q^2} \right) + \alpha \omega, \quad (33)$$

where

$$\omega = \left(\sqrt{p_0(p_0 + \sigma)(1 + \tau)} + \sqrt{p_0(p_0 - \sigma)(1 - \tau)} - 2p_0 \right)$$

Thus,

$$T_0 = q \left[\sqrt{1 + \frac{(1 - \alpha)^2 \tau^2 p_0^2}{\alpha^2 \omega^2}} - \frac{(1 - \alpha)\tau p_0}{\alpha \omega} \right]. \quad (34)$$

A finite pool trades off the fee revenue against the picking off risk, else pool size is infinite. It also has to be large enough so that $T_0 > q$. Clearly, $T_0 > q$, if and only if $\frac{(1 - \alpha)\tau p_0}{\alpha \omega} < 0$, which holds if $\omega < 0$.

Omega reaches a maximum value at zero.

$$\begin{aligned}
\frac{d\omega}{d\tau} &= \frac{\sqrt{p_0(p_0 + \sigma)}}{2\sqrt{(1 + \tau)}} - \frac{\sqrt{p_0(p_0 - \sigma)}}{2\sqrt{1 - \tau}}, \\
\frac{d^2\omega}{d\tau^2} &= -\frac{\sqrt{p_0(p_0 + \sigma)}}{4(1 + \tau)^{3/2}} - \frac{\sqrt{p_0(p_0 - \sigma)}}{2(1 - \tau)^{3/2}} < 0
\end{aligned}$$

$$\begin{aligned}\frac{d\omega}{d\tau} &= 0 \\ \implies \tau &= \frac{\sigma}{p_0}.\end{aligned}$$

At $\tau = \frac{\sigma}{p_0}$, $\omega = 0$. Thus, ω is strictly negative for $\tau \neq \frac{\sigma}{p_0}$

Further, to ensure that arbitrageur finds it profitable to trade, from the previous lemmas, $\tau \leq \frac{\sigma}{p_0}$.

■

Proof of Corollary 7

[i.] Immediate.

[ii.]

$$\frac{dT_0}{d\omega} = q \left[\frac{(1-\alpha)p_0\tau}{\alpha\omega^2} - \frac{(1-\alpha)^2 p_0^2 \tau^2}{\alpha^2 \omega^3 \sqrt{\left(\frac{(1-\alpha)^2 p_0^2 \tau^2}{\alpha^2 \omega^2} + 1\right)}} \right] \quad (35)$$

Notice that

$$\frac{(1-\alpha)p_0\tau}{\alpha\omega^2} > \frac{(1-\alpha)^2 p_0^2 \tau^2}{\alpha^2 \omega^3 \sqrt{\left(\frac{(1-\alpha)^2 p_0^2 \tau^2}{\alpha^2 \omega^2} + 1\right)}},$$

so $\frac{dT_0}{d\omega} > 0$. Further,

$$\frac{d\omega}{d\sigma} = \frac{1}{2} \left[\frac{\sqrt{p_0(1+\tau)}}{\sqrt{(p_0+\sigma)}} - \frac{\sqrt{p_0(1-\tau)}}{\sqrt{(p_0-\sigma)}} \right]. \quad (36)$$

$\frac{d\omega}{d\sigma} < 0$ if $\tau < \frac{\sigma}{p_0}$.

[iii.]

$$\frac{dT_0}{d\alpha} = \frac{p_0\tau \left(1 + \frac{p_0\tau(1-\alpha)}{\sqrt{p_0^2(1-\alpha)^2\tau^2 + \alpha^2\omega^2}}\right)}{\alpha^2\omega} < 0,$$

as $\omega < 0$.

■

Proof of Proposition 8

The Limit Order Market:

If two traders are in the market, the liquidity trader buys at price $p_0 + \alpha\sigma$ and sells at $p_0 - \alpha\sigma$. The trading costs are therefore $\alpha\sigma$.

If there is a monopolist liquidity supplier, he places orders at $p_0 - \sigma$ and $p_0 + \sigma$. The transaction cost is therefore σ .

The probability that a limit order liquidity supplier is a monopolist is $\gamma^*(1 - \gamma^*)$, and the probability that the passive trader faces a monopolist when there are two liquidity suppliers is $2\gamma^*(1 - \gamma^*) = \eta$.

The expected cost is $(1 - \eta)\alpha\sigma + \eta\sigma$

In the AMM:

$$e_\ell^b = \frac{E_0 T_0}{T_0 - q} - E_0$$

Using the fact that $p_0 = \frac{E_0}{T_0}$, yields

$$e_\ell^b = \frac{p_0 T_0^2}{T_0 - q} - p_0 T_0$$

From Proposition 6, the equilibrium supply of Tokens is given by

$$T_0 = q \left[\sqrt{1 + \frac{(1-\alpha)^2\tau^2 p_0^2}{\alpha^2\omega^2}} - \frac{(1-\alpha)\tau p_0}{\alpha\omega} \right]$$

where $\omega = \sqrt{p_0(p_0 + \sigma)(1 + \tau)} + \sqrt{\frac{p_0(p_0 - \sigma)}{1 + \tau}} - 2p_0$ which yields

$$\begin{aligned}
&= \frac{p_0 \left(q \left[\sqrt{1 + \frac{(1-\alpha)^2 \tau^2 p_0^2}{\alpha^2 \omega^2}} - \frac{(1-\alpha) \tau p_0}{\alpha \omega} \right] \right)^2}{q \left[\sqrt{1 + \frac{(1-\alpha)^2 \tau^2 p_0^2}{\alpha^2 \omega^2}} - \frac{(1-\alpha) \tau p_0}{\alpha \omega} \right] - q} - p_0 \left(q \left[\sqrt{1 + \frac{(1-\alpha)^2 \tau^2 p_0^2}{\alpha^2 \omega^2}} - \frac{(1-\alpha) \tau p_0}{\alpha \omega} \right] \right) \\
&= p_0 q \left[\frac{\left[\sqrt{1 + \frac{(1-\alpha)^2 \tau^2 p_0^2}{\alpha^2 \omega^2}} - \frac{(1-\alpha) \tau p_0}{\alpha \omega} \right]^2}{\sqrt{1 + \frac{(1-\alpha)^2 \tau^2 p_0^2}{\alpha^2 \omega^2}} - \frac{(1-\alpha) \tau p_0}{\alpha \omega} - 1} - \left[\sqrt{1 + \frac{(1-\alpha)^2 \tau^2 p_0^2}{\alpha^2 \omega^2}} - \frac{(1-\alpha) \tau p_0}{\alpha \omega} \right] \right] \\
&= p_0 q \left[\frac{\sqrt{1 + \frac{(1-\alpha)^2 \tau^2 p_0^2}{\alpha^2 \omega^2}} - \frac{(1-\alpha) \tau p_0}{\alpha \omega}}{\sqrt{1 + \frac{(1-\alpha)^2 \tau^2 p_0^2}{\alpha^2 \omega^2}} - \frac{(1-\alpha) \tau p_0}{\alpha \omega} - 1} \right] \\
&= p_0 q \lambda^b
\end{aligned}$$

Here,

$$\lambda^b = \left[\frac{\sqrt{1 + \frac{(1-\alpha)^2 \tau^2 p_0^2}{\alpha^2 \omega^2}} - \frac{(1-\alpha) \tau p_0}{\alpha \omega}}{\sqrt{1 + \frac{(1-\alpha)^2 \tau^2 p_0^2}{\alpha^2 \omega^2}} - \frac{(1-\alpha) \tau p_0}{\alpha \omega} - 1} \right] > 1$$

In addition, the liquidity trader also pays the liquidity fee of τ per ETH. Thus, the total payment to buy q tokens is $p_0 q (1 + \tau) \lambda^b$. Or, the per unit cost of trading q tokens is $p_0 ((1 + \tau) \lambda^b - 1)$.

Similarly,

$$\begin{aligned}
e_\ell^s &= E_0 - \frac{E_0 T_0}{T_0 + q} \\
&= p_0 T_0 - \frac{p_0 T_0^2}{T_0 + q} \\
&= p_0 \left(q \left[\sqrt{1 + \frac{(1-\alpha)^2 \tau^2 p_0^2}{\alpha^2 \omega^2}} - \frac{(1-\alpha) \tau p_0}{\alpha \omega} \right] \right) - \frac{p_0 \left(q \left[\sqrt{1 + \frac{(1-\alpha)^2 \tau^2 p_0^2}{\alpha^2 \omega^2}} - \frac{(1-\alpha) \tau p_0}{\alpha \omega} \right] \right)^2}{\left(q \left[\sqrt{1 + \frac{(1-\alpha)^2 \tau^2 p_0^2}{\alpha^2 \omega^2}} - \frac{(1-\alpha) \tau p_0}{\alpha \omega} \right] \right) + q} \\
&= p_0 q \left[\sqrt{1 + \frac{(1-\alpha)^2 \tau^2 p_0^2}{\alpha^2 \omega^2}} - \frac{(1-\alpha) \tau p_0}{\alpha \omega} - \frac{\left(\sqrt{1 + \frac{(1-\alpha)^2 \tau^2 p_0^2}{\alpha^2 \omega^2}} - \frac{(1-\alpha) \tau p_0}{\alpha \omega} \right)^2}{\sqrt{1 + \frac{(1-\alpha)^2 \tau^2 p_0^2}{\alpha^2 \omega^2}} - \frac{(1-\alpha) \tau p_0}{\alpha \omega} + 1} \right] \\
&= p_0 q \left[\frac{\sqrt{1 + \frac{(1-\alpha)^2 \tau^2 p_0^2}{\alpha^2 \omega^2}} - \frac{(1-\alpha) \tau p_0}{\alpha \omega}}{\sqrt{1 + \frac{(1-\alpha)^2 \tau^2 p_0^2}{\alpha^2 \omega^2}} - \frac{(1-\alpha) \tau p_0}{\alpha \omega} + 1} \right] \\
&= p_0 q \lambda^s
\end{aligned}$$

where

$$\lambda^s = \left[\frac{\sqrt{1 + \frac{(1-\alpha)^2 \tau^2 p_0^2}{\alpha^2 \omega^2}} - \frac{(1-\alpha) \tau p_0}{\alpha \omega}}{\sqrt{1 + \frac{(1-\alpha)^2 \tau^2 p_0^2}{\alpha^2 \omega^2}} - \frac{(1-\alpha) \tau p_0}{\alpha \omega} + 1} \right] < 1$$

In addition, the traders has to pay $\tau p_0 q \lambda^s$, thus the revenue from selling per token is $p_0(\lambda^s(1 - \tau))$

Thus, the expected ETH cost of trading tokens is

$$= p_0 \left(\frac{(1 + \tau) \lambda^b}{2} - \frac{\lambda^s(1 - \tau)}{2} \right) \tag{37}$$

■

Proof of Proposition 9

i. We first show that the liquidity trader prefers the AMM in the absence of informed trading ($\alpha = 0$) and then show that the limit order market dominates for $\alpha = 1$. Therefore there must be a critical α^* for which the liquidity trader is indifferent between markets.

Let

$$\xi := \frac{(1 - \alpha)\tau p_0}{\alpha\omega}. \quad (38)$$

From the proof of Proposition 6 we know that $\omega < 0$ if $\tau \neq \frac{\sigma}{p_0}$. It follows immediately that

$$\lim_{\alpha \rightarrow 0} \xi = -\infty \quad (39)$$

Then the poolsize $\zeta := \sqrt{1 + \xi^2} - \xi$ goes to positive infinity. To see this note that $\sqrt{\xi^2}$ and $-\xi$ are positive and go to infinity:

$$\lim_{\alpha \rightarrow 0} \zeta = \infty \quad (40)$$

We can then derive

$$\lim_{\alpha \rightarrow 0} \lambda^b = \lim_{\alpha \rightarrow 0} \frac{\zeta}{\zeta - 1} = \lim_{\alpha \rightarrow 0} \frac{1}{1 - \frac{1}{\zeta}} = 1 \quad (41)$$

and

$$\lim_{\alpha \rightarrow 0} \lambda^s = \lim_{\alpha \rightarrow 0} \frac{\zeta}{\zeta + 1} = \lim_{\alpha \rightarrow 0} \frac{1}{1 + \frac{1}{\zeta}} = 1 \quad (42)$$

The trading cost on the AMM as defined in Equation 37 then converges to

$$\lim_{\alpha \rightarrow 0} p_0 \left(\frac{(1 + \tau)\lambda^b}{2} - \frac{\lambda^s(1 - \tau)}{2} \right) = p_0 \frac{1 + \tau - (1 - \tau)}{2} = p_0 \tau \quad (43)$$

The trading cost in the limit order market when $\alpha = 0$ is $\eta\sigma$. In the absence of informed trading the AMM is thus cheaper for the liquidity trader whenever

$$\frac{\tau}{\eta} < \frac{\sigma}{p_0}. \quad (44)$$

To examine the price impact as $\alpha \rightarrow 1$, note that

$$\lim_{\alpha \rightarrow 1} \xi = 0 \quad (45)$$

Then the poolsize $\zeta = \sqrt{1 + \xi^2} - \xi$ goes to 1 and

$$\lim_{\alpha \rightarrow 1} \lambda^b = \lim_{\alpha \rightarrow 1} \frac{\xi}{\xi - 1} = \infty \quad (46)$$

Since we know that $0 < \lambda^s < 1$ we know that the trading cost on the AMM goes to

infinity:

$$\lim_{\alpha \rightarrow 1} p_0 \left(\frac{(1 + \tau)\lambda^b}{2} - \frac{\lambda^s(1 - \tau)}{2} \right) = \infty \quad (47)$$

The price impact on the limit order market when $\alpha = 1$ is simply σ . As $\alpha \rightarrow 1$ the limit order market is always the better trading venue for the liquidity trader.

ii. Price impact in the limit order market varies because the liquidity trader might face a monopolist or competitive market makers. The price impact on the AMM is deterministic.

■

INTERNET APPENDIX

B Data

Blockchain data is publicly available. Anyone can download and run an Ethereum node which collects all blockchain transactions. Transactions and interactions with smart contracts such as Uniswap leave log entries on the blockchain which can be accessed through an API from the Ethereum node or bought from a data provider. At the time of writing the Ethereum blockchain has about 3.8 billion log entries. We bought access to raw Ethereum log entries from Covalent. Not all data providers are of equal quality. Aggregate data such as periodic snapshots can miss important information as Dex liquidity can change substantially within a block or even within a transaction. Such intra-block changes will be missed by snapshots but captured by our log level data as we can reconstruct all transactions within the block in the sequence that they are executed. We are confident in our data as we did many robustness checks and comparisons with data directly obtained from our own Ethereum archive node.

Smart contracts are essentially small computer programs that execute in the Ethereum Virtual Machine. The code can generate log events and store the event and some data on the blockchain. The Uniswap smart contract generates several events every time a user interacts with it to deposit, withdraw, trade tokens or when a new pool is created. We filter over 200 million relevant log events and reverse engineer from the data that is stored with these log events how users interacted with the smart contracts.

Figure 15 shows one transaction in our sample¹⁵ on Etherscan, a commonly used web interface to examine Ethereum transactions. We see the transaction hash, that it was mined in block 13,738,106 on Dec 4th, 2021 at 5:47:46 UTC time. The user offered a gas price of 492.5 Gwei. Figure 16 shows the relevant log entries. The swap log event shows the two involved parties (one can be a router contract) and the amounts of tokens that flow in and out of the pool. The associated sync event shows the balance for each token in the liquidity pool. We carefully match swap and sync events as well as the minting and burning of liquidity tokens. Log events are not always recorded in the same sequence.

C Measuring volume

Dollar volume is a natural volume measure on US exchanges, and there is typically a recognized common price. On Uniswap, the US dollar is not the numeraire; most of the trading in the Uniswap system is against WETH or USD stablecoins. Further, the

¹⁵0xa3a7390e7de6587d05f17dbcac6911cd979b2d5ee00e3c361096d6ac8c93c77f

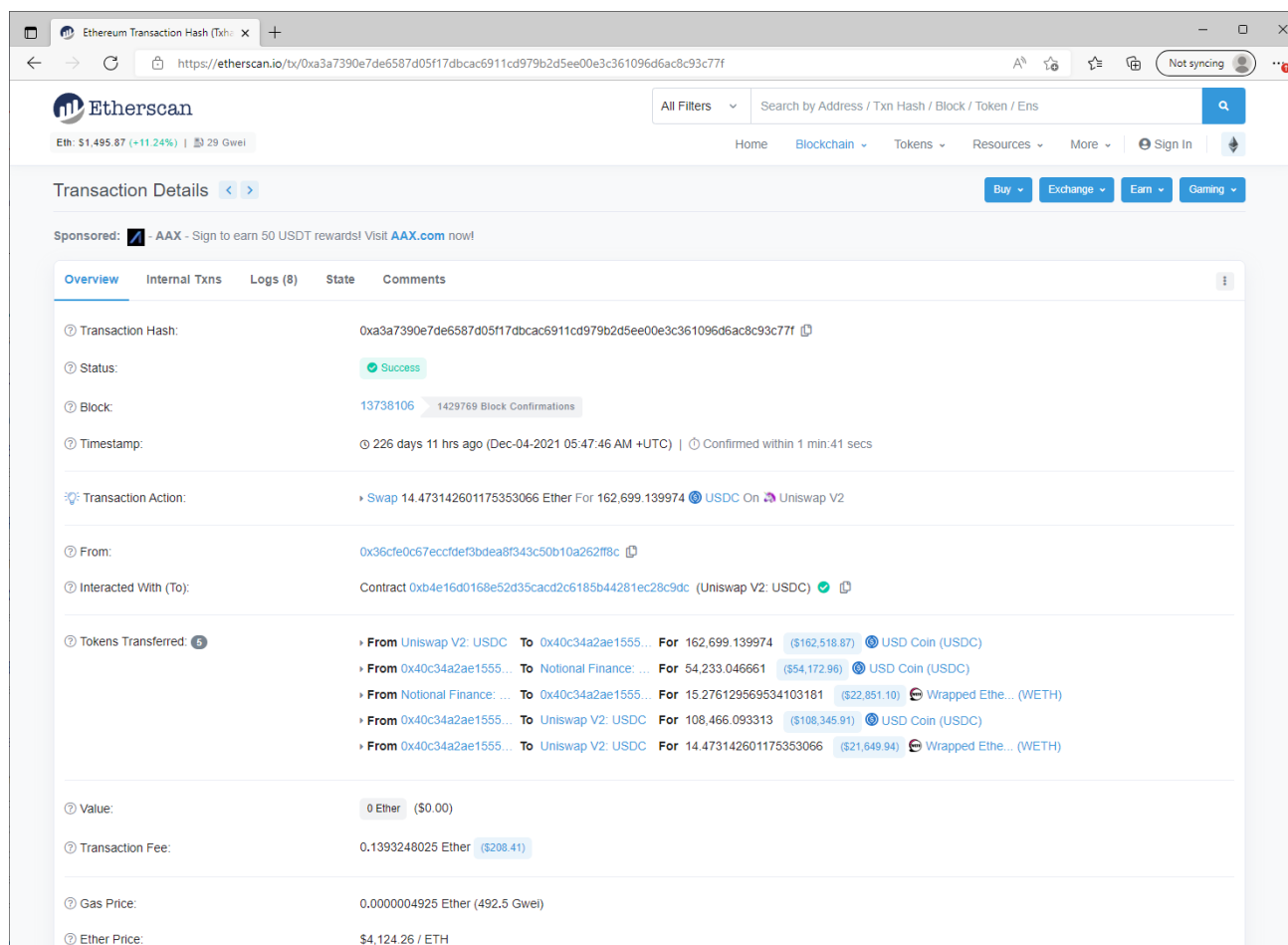


Figure 15. Example swap transaction

existence of flash loans and flash swaps during so called oracle attacks mean that there can be large price discrepancies across trading venues.

To compute volume, we take the WETH part of the trade and convert it to USD using Binance minute by minute data. We use the Binance price because the internal Uniswap price is sometimes purposely distorted. As we have indicated, large trades such as oracle attacks can push relative token prices out of equilibrium. Most of the remaining pools trade against a USD stablecoin. For those pools we convert stablecoin to USD. For all remaining pools we search for all pools where one of the tokens trades against WETH and convert using the prices from the pool with the highest volume. We then use external data from Binance to convert the ETH to USD.

As a concrete example of price distortions caused by oracle attacks, a December 17, 2020

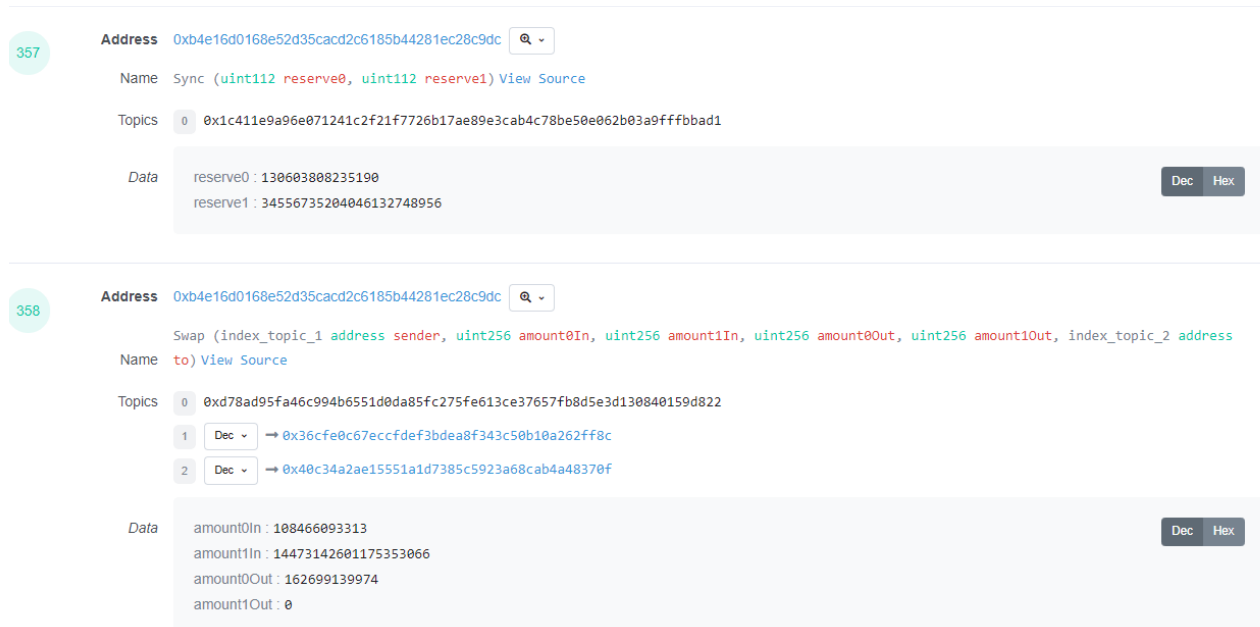


Figure 16. Example swap transaction - log entries

trade against Warp Finance is in our sample.¹⁶ The initiator borrowed about 200 million USD in flash loans from Uniswap and dYdX to manipulate the DAI/ETH price by trading 48.58 million DAI against 342,252 WETH. DAI is a USD stablecoin so one side of the trade roughly corresponds to 48 million USD. One ETH was worth about USD 643 at the time, making the WETH side of the trade worth about 220 million USD. The relatively complex trade involved depositing DAI and WETH into a liquidity pool, then using the liquidity tokens as collateral in Warp Finance. Finally, using a large trade to distort the DAI WETH price which led the Warp code to overvalue the deposited collateral.

Smart contract nuances aside, this trade was designed to distort the DAI/WETH price which means that using contemporaneous prices may distort the volume. Thus in the example above we would measure the volume as 220 million USD. Our approach does not systematically inflate volume because oracle attacks can happen in both directions and all trades that push prices out of equilibrium also have an opposing trade that brings prices back to equilibrium.

Many other contracts such as lending platforms rely on decentralized exchanges as price feeds or oracles to determine, for example, the value of the collateral in relation to the face value of an outstanding loan. Attackers can exploit poorly written code of such lending platforms for financial gain. Typically large trades are used to move prices in the bonding

¹⁶See transaction 0x8bb8dc5c7c830bac85fa48acad2505e9300a91c3ff239c9517d0cae33b595090.

curve market that the lending contract uses as oracle, making the smart contract believe that the collateral is very valuable. Then the attacker borrows against the collateral, brings the price on the bonding curve market back to equilibrium, and walks away from the, now under-collateralized, loan. One such attack happened on Harvest finance, a yield farming cooperative similar to a floating NAV money market fund. User could deposit tokens with Harvest finance in return for fAsset tokens (e.g. depositors of USDC receive a fUSDC token). The underlying tokens were invested in high yielding liquidity pools and the revenues are shared with the holders of fAsset tokens.

On October 26 an attacker borrowed 18 million USDT and 50 million USDC on Uniswap and converted over 17 million USDT to USDC on curve.fi (an automated market maker similar to Uniswap that specialized in trading stablecoins).¹⁷ This temporarily increased the price of USDC in the curve.fi pool, which is used as a price oracle for Harvest finance. The attacker then deposited USDC with Harvest, in exchange for an inflated number of fUSDC tokens. Specifically, the price manipulation caused the price of fUSDC to temporarily drop to 0.9712 from 0.98 before the attack. Then the attacker changed 17 million USDC back to USDT on curve.fi and sold his fUSDC tokens back to Harvest finance at 0.9833 as Harvest finance's smart contract updated the price based on the new information from curve.fi. The net profit of this attack was 619,408 USDC. The hacker then repeated the process 17 times and also attacked other Harvest finance pools for a total profit of USD 24 million.¹⁸

Transactions are finalized if they are incorporated onto the Ethereum blockchain, and anything that happens in the same transaction effectively happens at the same time. For this reason, flash swaps were introduced in Uniswap V2. This feature allows a user to borrow any amount up to the total liquidity available in a pool, only if the whole sum gets returned in the same Ethereum transaction. Because an Ethereum transaction is atomic, – i.e., it is either executed in its entirety or not at all – there is no credit risk for lenders as the loan is both originated and repaid in the same transaction.¹⁹ In our data, 735,644 transactions combine liquidity additions or removals with swaps or flash swaps. To be consistent with websites like uniswap.info we include flash swaps in volume computations in this paper. Liquidity providers earn a fee identical to the one on regular token swaps that is based on the gross amount of the flash loan regardless whether the repayment is the same token that was borrowed or not. For liquidity providers flash swaps offer a risk free way to earn higher fees.

Data Issues:

¹⁷see transaction 0x35f8d2f572fcea9c9288e5d462117850ef2694786992a8c3f6d02612277b0877.

¹⁸See 'Harvest Flashloan Economic Attack Post-Mortem', medium.com, Oct 26, 2020. <https://medium.com/harvest-finance/harvest-flashloan-economic-attack-post-mortem-3cf900d65217>

¹⁹Other protocols also offer flash loans, but Uniswap is unique because the borrowed amount can be repaid in any combination of pool tokens as long as the value repaid equals the value borrowed. Borrowers pay a fee on amounts borrowed.

We also observe some transactions that trade tokens back and forth without apparent reason. For example in one transaction on March 30, 2021 somebody created a new Uniswap exchange for a token named SCAMMY, borrowed WETH worth 220 million USD in a flash loan on dYdX, injected half as liquidity to the pool, and then traded 50 times the other half of the funds back and forth for a total volume of USD 5 billion.²⁰ The trader then withdrew the liquidity and repaid the flashloan. There is no obvious profit motive for this transaction. One possibility is that the trader tried to generate high fee revenue to place the token in a leading position at one of the yield farming websites in order to attract investment to this scam token (although naming the token scammy is not helpful for this purpose). We include such events in graphs and summary statistics. For our econometric analysis we winsorize the data and thus eliminate such outliers.

Ticker symbols on Uniswap are not protected. Anyone can create a token and assign the ticker symbol of a popular token like WETH or USDC. Tokens are uniquely identified by their address, e.g., 0xc02aaa39b223fe8d0a0e5c4f27ead9083c756cc2 for WETH, which is not easy to work with. Most people therefore use tickers and are exposed to copycat tokens. Table 6 lists fraudulent versions of popular tokens. We can see that, for example, Yearn Finance (YFI) has 17 copycat tokens that use the same ticker symbol. A total of 330 transactions were done on Uniswap with copycat tokens which is small compared to 284,973 transactions in the legitimate token. Overall we find that there amount of trading in fake tokens is small and will not affect our findings.

D Stylized facts on DEXs

In contrast to traditional exchanges, UniSwap liquidity pools are not certified. There are no listing requirements. A consequence of this is that some of the token pools are purposely misleading. For example, five different tokens in our sample share the same ticker symbol USDC. A naive user, who does not verify the relevant smart contract addresses, could be tricked into buying a worthless coin with the same ticker.²¹ Overall trading activity in these fraudulent pools is limited and will not affect our results. We provide detailed information on fake tokens in Appendix C.

Figure 17 presents the network of pools between all tokens that are part of the 50 largest pools by volume. The thickness of each edge corresponds to the trading volume between the tokens. We can see that Wrapped Ether (WETH) takes a central position in the Uniswap network. For our whole sample of 101,884 tokens in 105,098 pools we find that

²⁰See transaction 0xa8c00a56cf2455241bbc4b5ef9e3f9e761cdbc7909847ab8274dcd9bd1dded6a.

²¹The ‘real’ USDC stable coin resides under address 0xa0b86991c6218b36c1d19d4a2e9eb0ce3606eb48 and has over 4 million token transactions on the Ethereum blockchain. A token with the same ticker is 0x0xEFb9326678757522Ae4711d7fB5Cf321D6B664e6. Somebody created a Uniswap liquidity pool for this copycat token at the address 0x1bffb8a3fede9f83a3adc292ebf1716d40b220c1, which has a total of 10 trades and the size of the pool never exceeded 50 ETH.

Ticker	Number of fraudulent tokens	Fraudulent transactions	Nonfraudulent transactions
DAI	2	186,823	2,843,552
PAID	26	77,963	101,300
AXN	3	56,817	60,254
TRU	5	44,975	187,048
OM	11	42,830	89,406
BOND	24	41,637	133,864
DRC	6	41,519	50,802
PRQ	1	38,431	129,339
ANT	7	35,683	44,392
OCEAN	3	31,260	109,928
PBR	7	28,704	34,124
ORN	4	27,627	64,860
ALEPH	2	25,877	35,300
DUCK	13	24,562	77,409
YLD	13	24,113	63,087

Table 6. Fraudulent tokens. The table shows the number of fraudulent tokens, i.e. tokens with the same ticker symbol as popular tokens but with a different address. *Number of fraudulent tokens* is the number of fraudulent tokens found as part of a Uniswap liquidity pool. *Fraudulent transactions* are the number of transactions in liquidity pools with these fraudulent tokens. *Non-fraudulent transactions* is the number of transactions in the original token in Uniswap pools.

99,863 pools, or 95.02%, trade directly against ETH or WETH. The second highest number of tokens, 2,148, trade against USDT. The highest volume and the most connections are between WETH and USD stable coins such as USDT, USDC, and DAI. 5,244 tokens or 4.99% of tokens are trading directly against these three stablecoins. We note in passing that the Uniswap network has a core-periphery structure similar to many other financial networks. 99,491 tokens or 97.65% of tokens trade only against one other token.

E Numerical Example of an AMM

Assume that the fair exchange rate for a token is 10 ETH/token and a sole liquidity provider contributed $E = 100$ ETH and $T = 10$ tokens to the liquidity pool for which he gets 100 liquidity tokens in return. Suppose that the fee is $\tau = 0.003$.

Example 1 *A trader wants to buy tokens for $e = 10$ ETH. He gets $\frac{0.997eT}{0.997e+E} = \frac{0.997 \cdot 10 \cdot 10}{0.997 \cdot 10 + 100} = 0.90661$ tokens in return. The pool collects a fee of $0.003e = 0.003 \cdot 10 = 0.03$ ETH. The new token balance post trade is $10 - 0.90661 = 9.09339$ tokens.*

The post trade ETH balance equals the old balance plus what the trader gave for tokens

- (a) **True price is 10 ETH/token** *Had she kept her initial investment of 100 ETH and 10 tokens in a private wallet it would now be worth $100 + 10 \cdot 10 = 200$ ETH.*

When she redeems the liquidity token, she would obtain a total of $110 + 9.09339 \cdot 10 = 200.9339$ ETH and makes a profit of 0.9339 ETH. This is the sum of the trading fee ($\tau e = 0.003 \cdot 10 = 0.03$) and the gain from selling to the trader at an average price above the true price.

- (b) **True price is 12.0967 ETH/token** *Had she kept her initial investment of 100 ETH and 10 tokens in a private wallet it would now be worth $100 + 10 \cdot 12.0967 = 220.0967$*

When she redeems the liquidity token she gets $110 + 9.09339 \cdot 12.0967 = 220$. She loses 0.0967 ETH, in which the gain from the trading fee is more than offset by the loss from the exchange selling tokens at stale prices.

As with any passive liquidity provider, the Uniswap pools present a free option to the market. That is if the quantities in the pool are such that the terms of trade differ from the true value, arbitrageurs are more likely to pick off stale liquidity. This logic is reflected in the previous example. However, liquidity demanding trades are always valuable. The liquidity suppliers receive a fee for the liquidity demanding order and also receive a fee when equilibrium is replenished by arbitrage traders. They only face potential losses if there has been a permanent value change in the token.

If the price of a token moves away from the fundamental value because of a large order, an arbitrageur will initiate an offsetting trade and bring the mid-price of the exchange back to the fundamental value. Such short term deviations from the fundamental price of a token are beneficial to liquidity holders. In a pool without fees liquidity-providers will gain zero on such a trading pattern. Trades are always priced in such a way that the amount of ETH and tokens are on the bonding curve before and after any trade (see Equation 1). Thus a move from (E, T) to (E', T') and then back to (E, T) will leave the liquidity providers at exactly the same point they started from. Many crypto-traders refer to gains or losses while the pool is off equilibrium at value (E', T') as *impermanent loss*. With positive fees liquidity traders benefit from such short term deviations as they collect a proportional fee for both trades. In our empirical analysis we will estimate such short term deviations from a fundamental value as reversals.

Finally, even though arbitrageurs replenish the liquidity pool after a large, they pay a fee for doing it. This differs from a traditional limit order book in which liquidity is replenished by rivalrous liquidity suppliers.

Example 2 *Continue Example 1, case (a) and suppose that an arbitrageur brings back the price closer to the fundamental value. Assume that the arbitrageur can buy tokens at*

the fundamental value of 10, sells them to the pool at price $p(t)$, and chooses the optimum amount of tokens t to sell to the pool to maximize profit, $\pi = t(p(t) - 10)$. By sending t tokens to the pool he will obtain $e = \frac{(1-\tau)tE'}{T'+t(1-\tau)} = \frac{0.997 \cdot t \cdot 110}{9.09339 + 0.997t}$ ETH in return, resulting in a price $p(t) = e/t$. Solving for the optimal t that maximizes the arbitrageur's profit we find $t = 0.895648$ which is smaller than the amount of token sold by the pool in Example 1 because (i) the invariant k has changed after the first trade due to the fee revenue and (ii) fees make it optimal for the arbitrageur to sell a smaller amount back to the pool. It is easy to verify that without fees, i.e. $\tau = 0$, the invariant does not change after the first trade and the arbitrageur would sell exactly the same amount of tokens back to the pool that the pool sold in the previous trade, and the new mid-price of the pool would exactly equal the fundamental value. With fees, however, the arbitrageur optimally sells $t = 0.895648$ tokens to the pool for which he receives 9.836 ETH, leaving the pool with a new balance of 100.164 ETH and 9.98904 token. The new pool mid-price is 10.0274, which deviates slightly from the fundamental value of 10. Liquidity providers value their token holdings at the fundamental value of 10 and hold a total of $100.164 + 99.8904 = 200.054$ which is higher than their initial investment of 200.

F Gas Fees

Gas fees are the transaction costs on Ethereum blockchain. In the context of our paper gas fees are important for two reasons. First, they are one quantifiable component of the cost difference between centralized and decentralized exchanges. Second, gas costs that are high in times of market stress provide incentives for liquidity providers not to withdraw.

Gas is quoted in units of gwei, where 1 Gwei is 10^{-9} ETH. The amount of gas used for a specific transaction consists of a predictable component and a state contingent component. The predictable component comes from the organization of the Ethereum Virtual Machine. Any transaction comprises a known number of individual low-level computational steps. Fixed gas amounts are assigned to each type of operation and are independent of market conditions – the total gas used for a transaction is simply the sum of each computational step. In addition, users offer an amount of ETH per unit of gas consumed by the program, the so-called gas price, as an incentive to the miners to be included. The total gas fee is then the gas amount multiplied by the gas price. The gas price varies with market conditions and demand for transactions. Given a market price of Ethereum, the gas fee can be translated into USD.

We note three stylized facts that are relevant for our analysis. First, the average gas fee is low, second, the state contingent fee can be high and finally, the incremental fee relative to trading on Binance is low. Intuitively, the high state contingent fee acts as a pre-commitment not to withdraw liquidity in stressed market events.

To compare gas fees on a Cex and a Dex we compare a swap transaction to two tokens transfers, which would be required to move a token in and out of a centralized exchange. To measure gas costs we pick a sample of 200,000 blocks starting from block 11,668,958. Gas prices are recorded per transaction and each transaction can contain multiple swaps or multiple transfers or even other interactions with DeFi platforms. To generate a conservative estimate, we pick the 10 lowest gas amounts for transactions that contain swaps or transfers, respectively. These transactions are most likely to contain a ‘minimalist’ transaction that is, for example, just a simple token transfer. We manually verify on Etherscan that this is indeed the case. In these data, a Uniswap V2 swap uses on average 49.973 units of gas and a transfer uses 19.830 units of gas.²² A swap therefore uses on average 10.313 gas units more than the two token transfers that would be required to move tokens to and from a centralized exchange. Using a similar approach, we measure the gas amount of a liquidity withdrawal as 97,686 gas units. Liquidity withdrawals are the most expensive interactions with Uniswap as the smart contract has to compute and transfer the liquidity provider’s share of the fee revenue.

Using average gas prices can be misleading for two reasons. First, gas prices have a fat right tail and averages might be driven by outliers. Second, market conditions might make it very lucrative to use one DeFi protocol over another one at a given time or users might have different preferences to get their transaction included in a timely manner. To best capture preferences of Uniswap users, on each day we compute the median gas price for all Uniswap transactions, multiply that with the incremental gas amount of 10.312 gas units, and convert that to USD. Figure 18 illustrates our findings. The average incremental fee is 0.94 cents, the median incremental fee is 0.22 cents. On some days the incremental fee can be very high. Indeed, the maximum in our sample is USD 11.57 on May 11, 2021 when ETH first broke through USD 4,000. For the second part of our sample, since January 2021, the median incremental fee was USD 1.33.

Finally, observe that our calculation does not include any withdrawal fees that an exchange may charge. Binance, for example, charges token specific fees that are hard to include in a comparison. Binance charges a fee of 0.005 Eth for ETH withdrawals to the Ethereum blockchain, which at the time of writing the paper would be equivalent to USD 8.5.

²²We observe some very small variations in gas amounts due to minor differences in the implementation of ERC-20 token contracts.

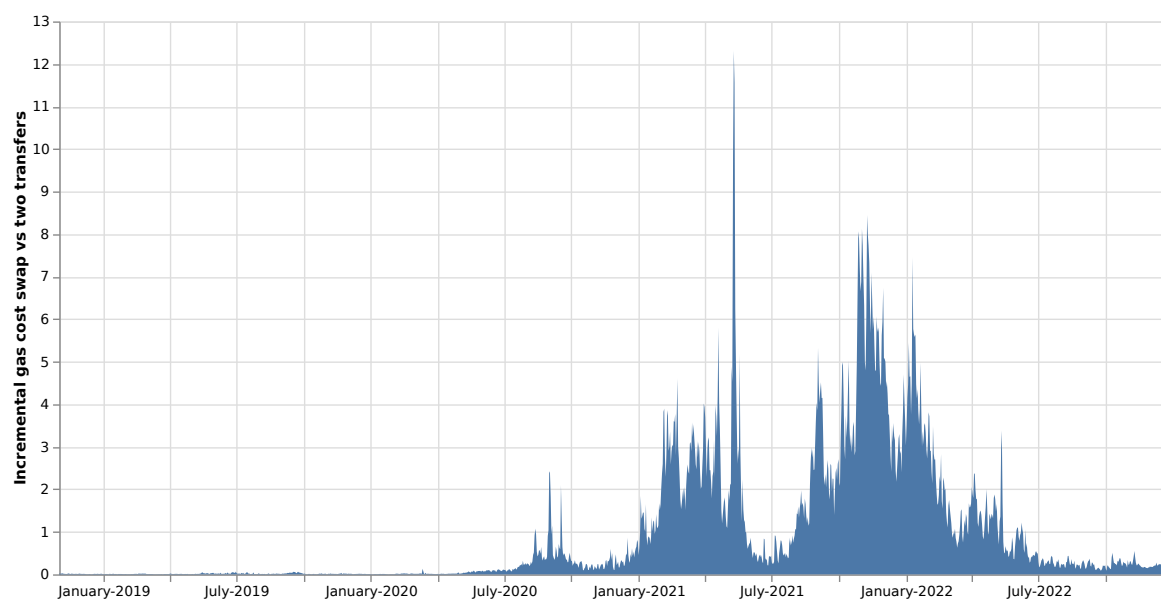


Figure 18. Incremental Fees of a swap transaction over two token transfers measured in USD.